

R-30*i*B Auxiliary Axis Option

MAINTENANCE MANUAL

MAREBAUXA07131E REV. A

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FANUC America Corporation Patent List

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Conventions

WARNING

Information appearing under the "WARNING" caption concerns the protection of personnel. It is boxed and bolded to set it apart from the surrounding text.

CAUTION

Information appearing under the "CAUTION" caption concerns the protection of equipment, software, and data. It is boxed and bolded to set it apart from the surrounding text.

Note Information appearing next to NOTE concerns related information or useful hints.

Safety

FANUC America Corporation is not and does not represent itself as an expert in safety systems, safety equipment, or the specific safety aspects of your company and/or its work force. It is the responsibility of the owner, employer, or user to take all necessary steps to guarantee the safety of all personnel in the workplace.

The appropriate level of safety for your application and installation can be best determined by safety system professionals. FANUC America Corporation therefore, recommends that each customer consult with such professionals in order to provide a workplace that allows for the safe application, use, and operation of FANUC America Corporation systems.

According to the industry standard ANSI/RIA R15-06, the owner or user is advised to consult the standards to ensure compliance with its requests for Robotics System design, usability, operation, maintenance, and service. Additionally, as the owner, employer, or user of a robotic system, it is your responsibility to arrange for the training of the operator of a robot system to recognize and respond to known hazards associated with your robotic system and to be aware of the recommended operating procedures for your particular application and robot installation.

Ensure that the robot being used is appropriate for the application. Robots used in classified (hazardous) locations must be certified for this use.

FANUC America Corporation therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the robotics system be trained in an approved FANUC America Corporation training course and become familiar with the proper operation of the system. Persons responsible for programming the system—including the design, implementation, and debugging of application programs—must be familiar with the recommended programming procedures for your application and robot installation.

The following guidelines are provided to emphasize the importance of safety in the workplace.

CONSIDERING SAFETY FOR YOUR ROBOT INSTALLATION

Safety is essential whenever robots are used. Keep in mind the following factors with regard to safety:

- The safety of people and equipment
- Use of safety enhancing devices
- Techniques for safe teaching and manual operation of the robot(s)
- Techniques for safe automatic operation of the robot(s)
- Regular scheduled inspection of the robot and workcell
- Proper maintenance of the robot

Keeping People Safe

The safety of people is always of primary importance in any situation. When applying safety measures to your robotic system, consider the following:

- External devices
- Robot(s)
- Tooling
- Workpiece

Using Safety Enhancing Devices

Always give appropriate attention to the work area that surrounds the robot. The safety of the work area can be enhanced by the installation of some or all of the following devices:

- Safety fences, barriers, or chains
- Light curtains
- Interlocks
- Pressure mats
- Floor markings
- Warning lights
- Mechanical stops
- EMERGENCY STOP buttons
- DEADMAN switches

Setting Up a Safe Workcell

A safe workcell is essential to protect people and equipment. Observe the following guidelines to ensure that the workcell is set up safely. These suggestions are intended to supplement and not replace existing federal, state, and local laws, regulations, and guidelines that pertain to safety.

- Sponsor your personnel for training in approved FANUC America Corporation training course(s) related to your application. Never permit untrained personnel to operate the robots.
- Install a lockout device that uses an access code to prevent unauthorized persons from operating the robot.
- Use anti-tie-down logic to prevent the operator from bypassing safety measures.
- Arrange the workcell so the operator faces the workcell and can see what is going on inside the cell.
- Clearly identify the work envelope of each robot in the system with floor markings, signs, and special barriers. The work envelope is the area defined by the maximum motion range of the robot, including any tooling attached to the wrist flange that extend this range.

- Position all controllers outside the robot work envelope.
- Never rely on software or firmware based controllers as the primary safety element unless they comply with applicable current robot safety standards.
- Mount an adequate number of EMERGENCY STOP buttons or switches within easy reach of the operator and at critical points inside and around the outside of the workcell.
- Install flashing lights and/or audible warning devices that activate whenever the robot is operating, that is, whenever power is applied to the servo drive system. Audible warning devices shall exceed the ambient noise level at the end-use application.
- Wherever possible, install safety fences to protect against unauthorized entry by personnel into the work envelope.
- Install special guarding that prevents the operator from reaching into restricted areas of the work envelope.
- Use interlocks.
- Use presence or proximity sensing devices such as light curtains, mats, and capacitance and vision systems to enhance safety.
- Periodically check the safety joints or safety clutches that can be optionally installed between the robot wrist flange and tooling. If the tooling strikes an object, these devices dislodge, remove power from the system, and help to minimize damage to the tooling and robot.
- Make sure all external devices are properly filtered, grounded, shielded, and suppressed to prevent hazardous motion due to the effects of electro-magnetic interference (EMI), radio frequency interference (RFI), and electro-static discharge (ESD).
- Make provisions for power lockout/tagout at the controller.
- Eliminate *pinch points*. Pinch points are areas where personnel could get trapped between a moving robot and other equipment.
- Provide enough room inside the workcell to permit personnel to teach the robot and perform maintenance safely.
- Program the robot to load and unload material safely.
- If high voltage electrostatics are present, be sure to provide appropriate interlocks, warning, and beacons.
- If materials are being applied at dangerously high pressure, provide electrical interlocks for lockout of material flow and pressure.

Staying Safe While Teaching or Manually Operating the Robot

Advise all personnel who must teach the robot or otherwise manually operate the robot to observe the following rules:

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Know whether or not you are using an intrinsically safe teach pendant if you are working in a hazardous environment.

- Before teaching, visually inspect the robot and work envelope to make sure that no potentially hazardous conditions exist. The work envelope is the area defined by the maximum motion range of the robot. These include tooling attached to the wrist flange that extends this range.
- The area near the robot must be clean and free of oil, water, or debris. Immediately report unsafe working conditions to the supervisor or safety department.
- FANUC America Corporation recommends that no one enter the work envelope of a robot that is on, except for robot teaching operations. However, if you must enter the work envelope, be sure all safeguards are in place, check the teach pendant DEADMAN switch for proper operation, and place the robot in teach mode. Take the teach pendant with you, turn it on, and be prepared to release the DEADMAN switch. Only the person with the teach pendant should be in the work envelope.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

- Know the path that can be used to escape from a moving robot; make sure the escape path is never blocked.
- Isolate the robot from all remote control signals that can cause motion while data is being taught.
- Test any program being run for the first time in the following manner:

 WARNING

Stay outside the robot work envelope whenever a program is being run. Failure to do so can result in injury.

- Using a low motion speed, single step the program for at least one full cycle.
 - Using a low motion speed, test run the program continuously for at least one full cycle.
 - Using the programmed speed, test run the program continuously for at least one full cycle.
- Make sure all personnel are outside the work envelope before running production.

Staying Safe During Automatic Operation

Advise all personnel who operate the robot during production to observe the following rules:

- Make sure all safety provisions are present and active.

- Know the entire workcell area. The workcell includes the robot and its work envelope, plus the area occupied by all external devices and other equipment with which the robot interacts.
- Understand the complete task the robot is programmed to perform before initiating automatic operation.
- Make sure all personnel are outside the work envelope before operating the robot.
- Never enter or allow others to enter the work envelope during automatic operation of the robot.
- Know the location and status of all switches, sensors, and control signals that could cause the robot to move.
- Know where the EMERGENCY STOP buttons are located on both the robot control and external control devices. Be prepared to press these buttons in an emergency.
- Never assume that a program is complete if the robot is not moving. The robot could be waiting for an input signal that will permit it to continue its activity.
- If the robot is running in a pattern, do not assume it will continue to run in the same pattern.
- Never try to stop the robot, or break its motion, with your body. The only way to stop robot motion immediately is to press an EMERGENCY STOP button located on the controller panel, teach pendant, or emergency stop stations around the workcell.

Staying Safe During Inspection

When inspecting the robot, be sure to

- Turn off power at the controller.
- Lock out and tag out the power source at the controller according to the policies of your plant.
- Turn off the compressed air source and relieve the air pressure.
- If robot motion is not needed for inspecting the electrical circuits, press the EMERGENCY STOP button on the operator panel.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the robot motion or electrical circuits, be prepared to press the EMERGENCY STOP button, in an emergency.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Staying Safe During Maintenance

When performing maintenance on your robot system, observe the following rules:

- Never enter the work envelope while the robot or a program is in operation.
- Before entering the work envelope, visually inspect the workcell to make sure no potentially hazardous conditions exist.

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Consider all or any overlapping work envelopes of adjoining robots when standing in a work envelope.
- Test the teach pendant for proper operation before entering the work envelope.
- If it is necessary for you to enter the robot work envelope while power is turned on, you must be sure that you are in control of the robot. Be sure to take the teach pendant with you, press the DEADMAN switch, and turn the teach pendant on. Be prepared to release the DEADMAN switch to turn off servo power to the robot immediately.
- Whenever possible, perform maintenance with the power turned off. Before you open the controller front panel or enter the work envelope, turn off and lock out the 3-phase power source at the controller.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

 WARNING

Lethal voltage is present in the controller WHENEVER IT IS CONNECTED to a power source. Be extremely careful to avoid electrical shock. HIGH VOLTAGE IS PRESENT at the input side whenever the controller is connected to a power source. Turning the disconnect or circuit breaker to the OFF position removes power from the output side of the device only.

- Release or block all stored energy. Before working on the pneumatic system, shut off the system air supply and purge the air lines.
- Isolate the robot from all remote control signals. If maintenance must be done when the power is on, make sure the person inside the work envelope has sole control of the robot. The teach pendant must be held by this person.
- Make sure personnel cannot get trapped between the moving robot and other equipment. Know the path that can be used to escape from a moving robot. Make sure the escape route is never blocked.
- Use blocks, mechanical stops, and pins to prevent hazardous movement by the robot. Make sure that such devices do not create pinch points that could trap personnel.

 WARNING

Do not try to remove any mechanical component from the robot before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.

- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.
- When replacing or installing components, make sure dirt and debris do not enter the system.
- Use only specified parts for replacement. To avoid fires and damage to parts in the controller, never use nonspecified fuses.
- Before restarting a robot, make sure no one is inside the work envelope; be sure that the robot and all external devices are operating normally.

KEEPING MACHINE TOOLS AND EXTERNAL DEVICES SAFE

Certain programming and mechanical measures are useful in keeping the machine tools and other external devices safe. Some of these measures are outlined below. Make sure you know all associated measures for safe use of such devices.

Programming Safety Precautions

Implement the following programming safety measures to prevent damage to machine tools and other external devices.

- Back-check limit switches in the workcell to make sure they do not fail.
- Implement “failure routines” in programs that will provide appropriate robot actions if an external device or another robot in the workcell fails.
- Use *handshaking* protocol to synchronize robot and external device operations.
- Program the robot to check the condition of all external devices during an operating cycle.

Mechanical Safety Precautions

Implement the following mechanical safety measures to prevent damage to machine tools and other external devices.

- Make sure the workcell is clean and free of oil, water, and debris.
- Use DCS (Dual Check Safety), software limits, limit switches, and mechanical hardstops to prevent undesired movement of the robot into the work area of machine tools and external devices.

KEEPING THE ROBOT SAFE

Observe the following operating and programming guidelines to prevent damage to the robot.

Operating Safety Precautions

The following measures are designed to prevent damage to the robot during operation.

- Use a low override speed to increase your control over the robot when jogging the robot.
- Visualize the movement the robot will make before you press the jog keys on the teach pendant.
- Make sure the work envelope is clean and free of oil, water, or debris.
- Use circuit breakers to guard against electrical overload.

Programming Safety Precautions

The following safety measures are designed to prevent damage to the robot during programming:

- Establish *interference zones* to prevent collisions when two or more robots share a work area.
- Make sure that the program ends with the robot near or at the home position.
- Be aware of signals or other operations that could trigger operation of tooling resulting in personal injury or equipment damage.
- In dispensing applications, be aware of all safety guidelines with respect to the dispensing materials.

NOTE: Any deviation from the methods and safety practices described in this manual must conform to the approved standards of your company. If you have questions, see your supervisor.

ADDITIONAL SAFETY CONSIDERATIONS FOR PAINT ROBOT INSTALLATIONS

Process technicians are sometimes required to enter the paint booth, for example, during daily or routine calibration or while teaching new paths to a robot. Maintenance personnel also must work inside the paint booth periodically.

Whenever personnel are working inside the paint booth, ventilation equipment must be used. Instruction on the proper use of ventilating equipment usually is provided by the paint shop supervisor.

Although paint booth hazards have been minimized, potential dangers still exist. Therefore, today's highly automated paint booth requires that process and maintenance personnel have full awareness of the system and its capabilities. They must understand the interaction that occurs between the vehicle moving along the conveyor and the robot(s), hood/deck and door opening devices, and high-voltage electrostatic tools.



CAUTION

Ensure that all ground cables remain connected. Never operate the paint robot with ground provisions disconnected. Otherwise, you could injure personnel or damage equipment.

Paint robots are operated in three modes:

- Teach or manual mode
- Automatic mode, including automatic and exercise operation
- Diagnostic mode

During both teach and automatic modes, the robots in the paint booth will follow a predetermined pattern of movements. In teach mode, the process technician teaches (programs) paint paths using the teach pendant.

In automatic mode, robot operation is initiated at the System Operator Console (SOC) or Manual Control Panel (MCP), if available, and can be monitored from outside the paint booth. All personnel must remain outside of the booth or in a designated safe area within the booth whenever automatic mode is initiated at the SOC or MCP.

In automatic mode, the robots will execute the path movements they were taught during teach mode, but generally at production speeds.

When process and maintenance personnel run diagnostic routines that require them to remain in the paint booth, they must stay in a designated safe area.

Paint System Safety Features

Process technicians and maintenance personnel must become totally familiar with the equipment and its capabilities. To minimize the risk of injury when working near robots and related equipment, personnel must comply strictly with the procedures in the manuals.

This section provides information about the safety features that are included in the paint system and also explains the way the robot interacts with other equipment in the system.

The paint system includes the following safety features:

- Most paint booths have red warning beacons that illuminate when the robots are armed and ready to paint. Your booth might have other kinds of indicators. Learn what these are.

- Some paint booths have a blue beacon that, when illuminated, indicates that the electrostatic devices are enabled. Your booth might have other kinds of indicators. Learn what these are.
- EMERGENCY STOP buttons are located on the robot controller and teach pendant. Become familiar with the locations of all E–STOP buttons.
- An intrinsically safe teach pendant is used when teaching in hazardous paint atmospheres.
- A DEADMAN switch is located on each teach pendant. When this switch is held in, and the teach pendant is on, power is applied to the robot servo system. If the engaged DEADMAN switch is released or pressed harder during robot operation, power is removed from the servo system, all axis brakes are applied, and the robot comes to an EMERGENCY STOP. Safety interlocks within the system might also E–STOP other robots.

 WARNING

An EMERGENCY STOP will occur if the DEADMAN switch is released on a bypassed robot.

- Overtravel by robot axes is prevented by software limits. All of the major and minor axes are governed by software limits. DCS (Dual Check Safety), limit switches and hardstops also limit travel by the major axes.
- EMERGENCY STOP limit switches and photoelectric eyes might be part of your system. Limit switches, located on the entrance/exit doors of each booth, will EMERGENCY STOP all equipment in the booth if a door is opened while the system is operating in automatic or manual mode. For some systems, signals to these switches are inactive when the switch on the SOC is in teach mode.
- When present, photoelectric eyes are sometimes used to monitor unauthorized intrusion through the entrance/exit silhouette openings.
- System status is monitored by computer. Severe conditions result in automatic system shutdown.

Staying Safe While Operating the Paint Robot

When you work in or near the paint booth, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.

 WARNING

Observe all safety rules and guidelines to avoid injury.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

 WARNING

Enclosures shall not be opened unless the area is known to be nonhazardous or all power has been removed from devices within the enclosure. Power shall not be restored after the enclosure has been opened until all combustible dusts have been removed from the interior of the enclosure and the enclosure purged. Refer to the Purge chapter for the required purge time.

- Know the work area of the entire paint station (workcell).
- Know the work envelope of the robot and hood/deck and door opening devices.
- Be aware of overlapping work envelopes of adjacent robots.
- Know where all red, mushroom-shaped EMERGENCY STOP buttons are located.
- Know the location and status of all switches, sensors, and/or control signals that might cause the robot, conveyor, and opening devices to move.
- Make sure that the work area near the robot is clean and free of water, oil, and debris. Report unsafe conditions to your supervisor.
- Become familiar with the complete task the robot will perform BEFORE starting automatic mode.
- Make sure all personnel are outside the paint booth before you turn on power to the robot servo system.
- Never enter the work envelope or paint booth before you turn off power to the robot servo system.
- Never enter the work envelope during automatic operation unless a safe area has been designated.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Remove all metallic objects, such as rings, watches, and belts, before entering a booth when the electrostatic devices are enabled.
- Stay out of areas where you might get trapped between a moving robot, conveyor, or opening device and another object.
- Be aware of signals and/or operations that could result in the triggering of guns or bells.
- Be aware of all safety precautions when dispensing of paint is required.
- Follow the procedures described in this manual.

Special Precautions for Combustible Dusts (Powder Paint)

When the robot is used in a location where combustible dusts are found, such as the application of powder paint, the following special precautions are required to insure that there are no combustible dusts inside the robot.

- Purge maintenance air should be maintained at all times, even when the robot power is off. This will insure that dust can not enter the robot.
- A purge cycle will not remove accumulated dusts. Therefore, if the robot is exposed to dust when maintenance air is not present, it will be necessary to remove the covers and clean out any accumulated dust. Do not energize the robot until you have performed the following steps.
 1. Before covers are removed, the exterior of the robot should be cleaned to remove accumulated dust.
 2. When cleaning and removing accumulated dust, either on the outside or inside of the robot, be sure to use methods appropriate for the type of dust that exists. Usually lint free rags dampened with water are acceptable. Do not use a vacuum cleaner to remove dust as it can generate static electricity and cause an explosion unless special precautions are taken.
 3. Thoroughly clean the interior of the robot with a lint free rag to remove any accumulated dust.
 4. When the dust has been removed, the covers must be replaced immediately.
 5. Immediately after the covers are replaced, run a complete purge cycle. The robot can now be energized.

Staying Safe While Operating Paint Application Equipment

When you work with paint application equipment, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.



WARNING

When working with electrostatic paint equipment, follow all national and local codes as well as all safety guidelines within your organization. Also reference the following standards: NFPA 33 Standards for Spray Application Using Flammable or Combustible Materials, and NFPA 70 National Electrical Code.

- **Grounding:** All electrically conductive objects in the spray area must be grounded. This includes the spray booth, robots, conveyors, workstations, part carriers, hooks, paint pressure pots, as well as solvent containers. Grounding is defined as the object or objects shall be electrically connected to ground with a resistance of not more than 1 megohms.
- **High Voltage:** High voltage should only be on during actual spray operations. Voltage should be off when the painting process is completed. Never leave high voltage on during a cap cleaning process.
- Avoid any accumulation of combustible vapors or coating matter.
- Follow all manufacturer recommended cleaning procedures.
- Make sure all interlocks are operational.

- No smoking.
- Post all warning signs regarding the electrostatic equipment and operation of electrostatic equipment according to NFPA 33 Standard for Spray Application Using Flammable or Combustible Material.
- Disable all air and paint pressure to bell.
- Verify that the lines are not under pressure.

Staying Safe During Maintenance

When you perform maintenance on the painter system, observe the following rules, and all other maintenance safety rules that apply to all robot installations. Only qualified, trained service or maintenance personnel should perform repair work on a robot.

- Paint robots operate in a potentially explosive environment. Use caution when working with electric tools.
- When a maintenance technician is repairing or adjusting a robot, the work area is under the control of that technician. All personnel not participating in the maintenance must stay out of the area.
- For some maintenance procedures, station a second person at the control panel within reach of the EMERGENCY STOP button. This person must understand the robot and associated potential hazards.
- Be sure all covers and inspection plates are in good repair and in place.
- Always return the robot to the “home” position before you disarm it.
- Never use machine power to aid in removing any component from the robot.
- During robot operations, be aware of the robot’s movements. Excess vibration, unusual sounds, and so forth, can alert you to potential problems.
- Whenever possible, turn off the main electrical disconnect before you clean the robot.
- When using vinyl resin observe the following:
 - Wear eye protection and protective gloves during application and removal.
 - Adequate ventilation is required. Overexposure could cause drowsiness or skin and eye irritation.
 - If there is contact with the skin, wash with water.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.
- When using paint remover observe the following:
 - Eye protection, protective rubber gloves, boots, and apron are required during booth cleaning.
 - Adequate ventilation is required. Overexposure could cause drowsiness.
 - If there is contact with the skin or eyes, rinse with water for at least 15 minutes. Then seek medical attention as soon as possible.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.

R-30iB Auxiliary Axis Option Maintenance Manual

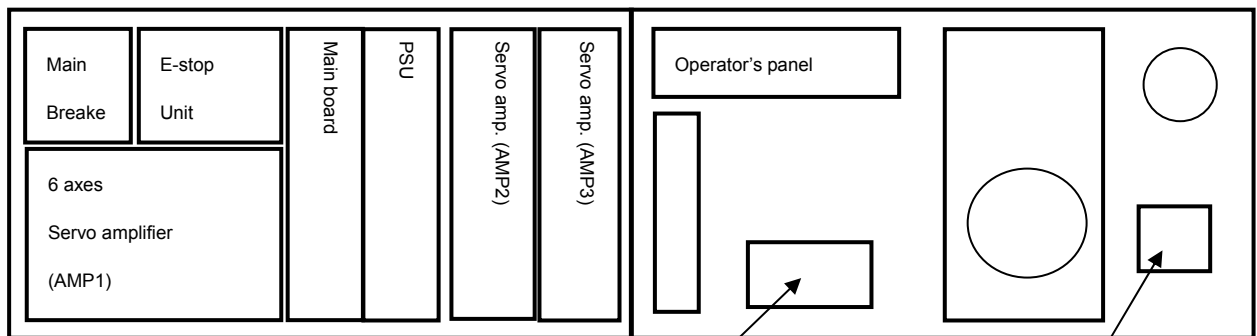
1. Scope

This manual is applied for Auxiliary Axis Option of the R-30iB controller (Standard, NRTL and CE). Refer to the R-30iB maintenance manual for general maintenance description.

2. Assembly

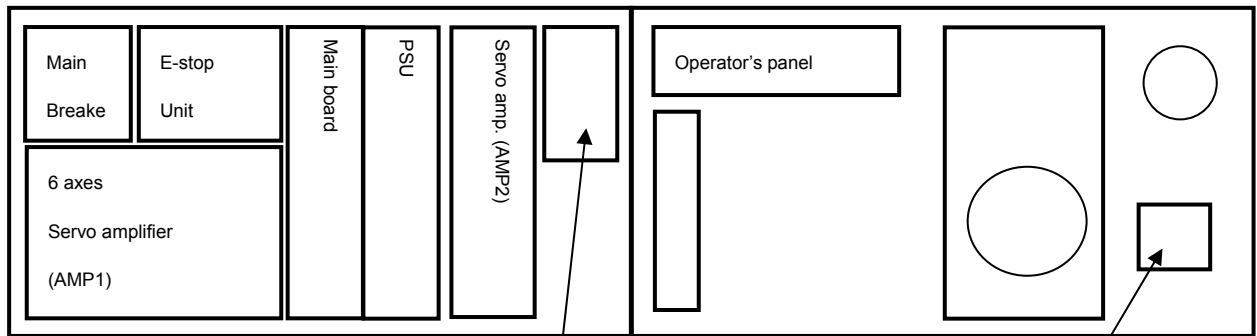
2.1. A-cabinet

2.1.1. In case of Resistor Discharge



Brake unit/Gun change unit

Battery unit
C size alkali dry cells
Quantity: 4

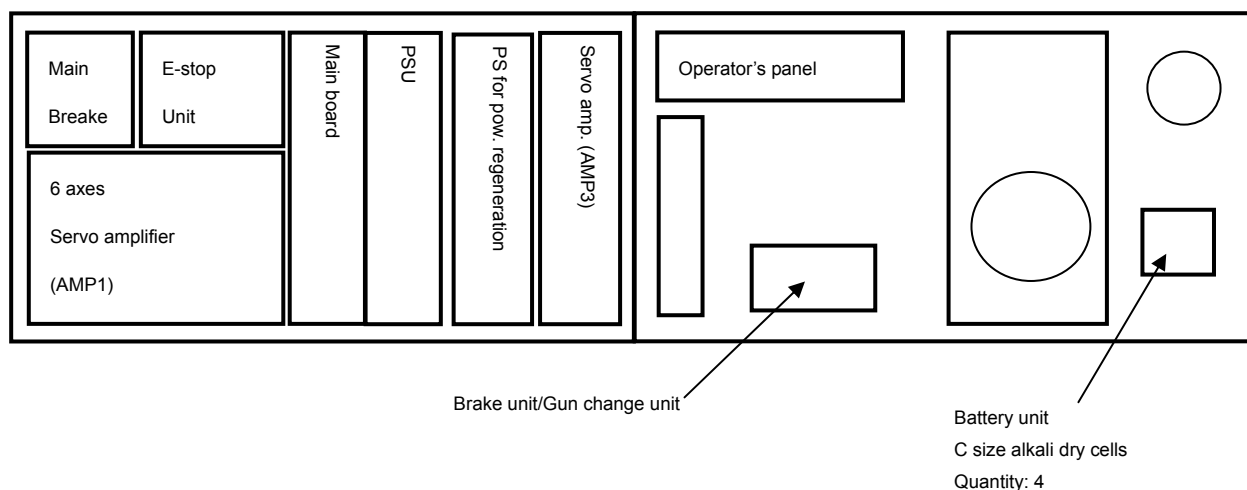


Brake unit

Battery unit
C size alkali dry cells
Quantity: 4

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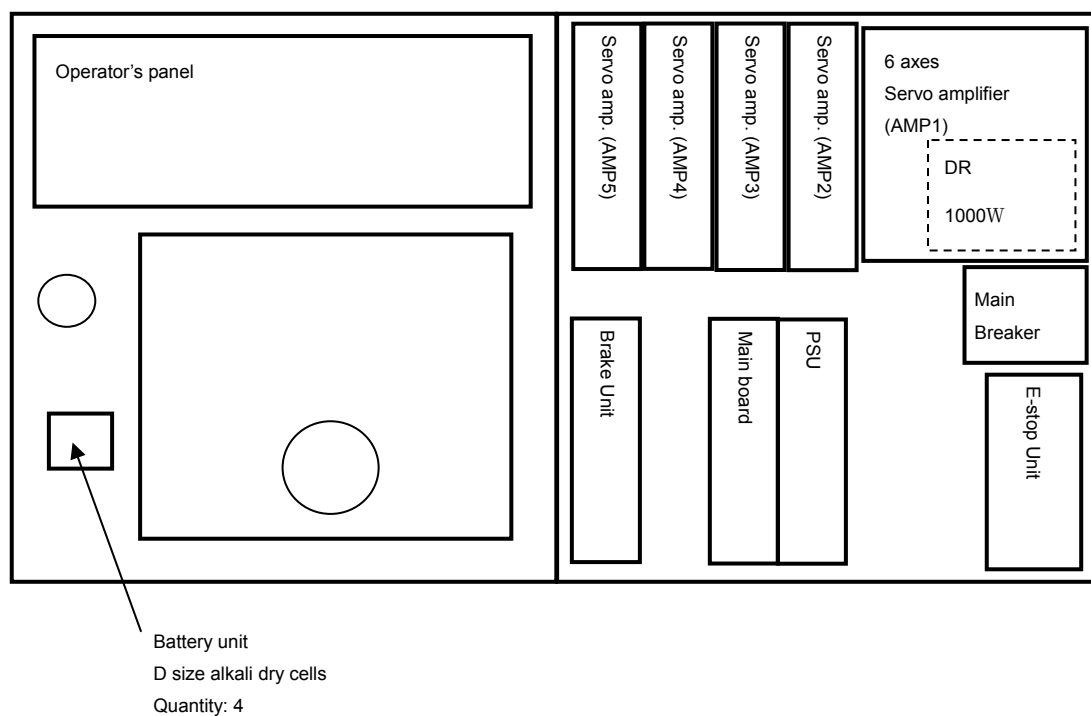
2.1.2. In case of Power regeneration



2.2. B-cabinet

2.2.1. In case of Discharge resistor A(1 element/1000W, A05B-2603-H060)

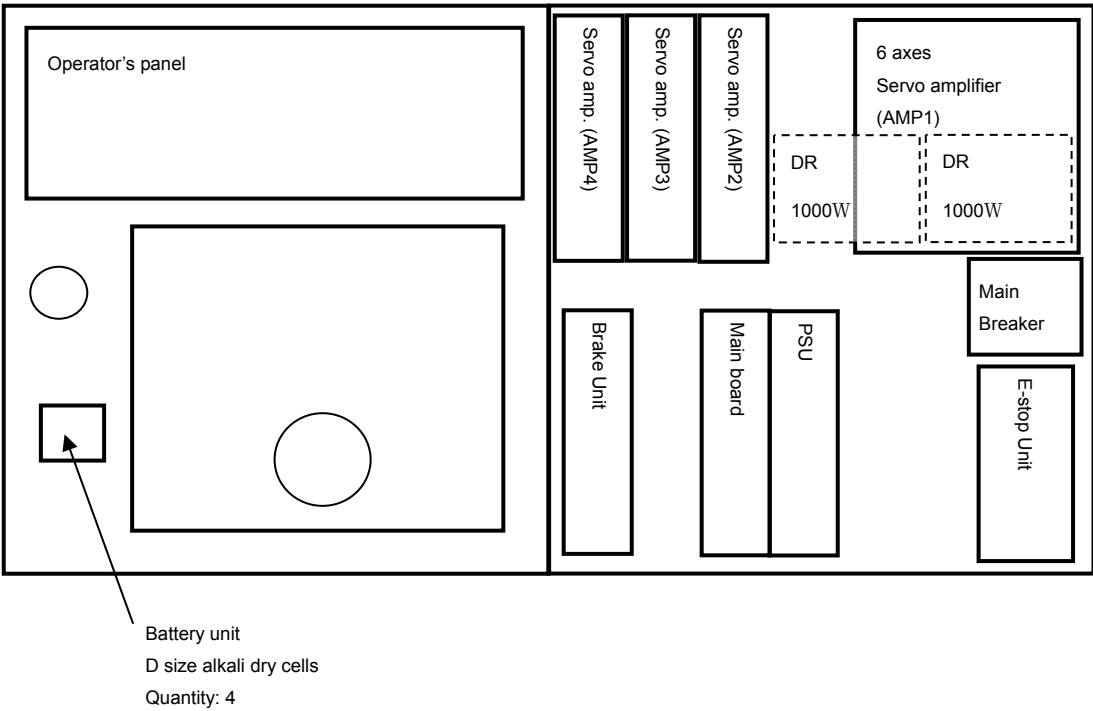
Robot: R-2000iB, R-1000iA, M-710iC, M-420iA, ARC Mate 100iC, ARC Mate 120iC, M-10iA, M-20iA, M-430iA, F-200iB



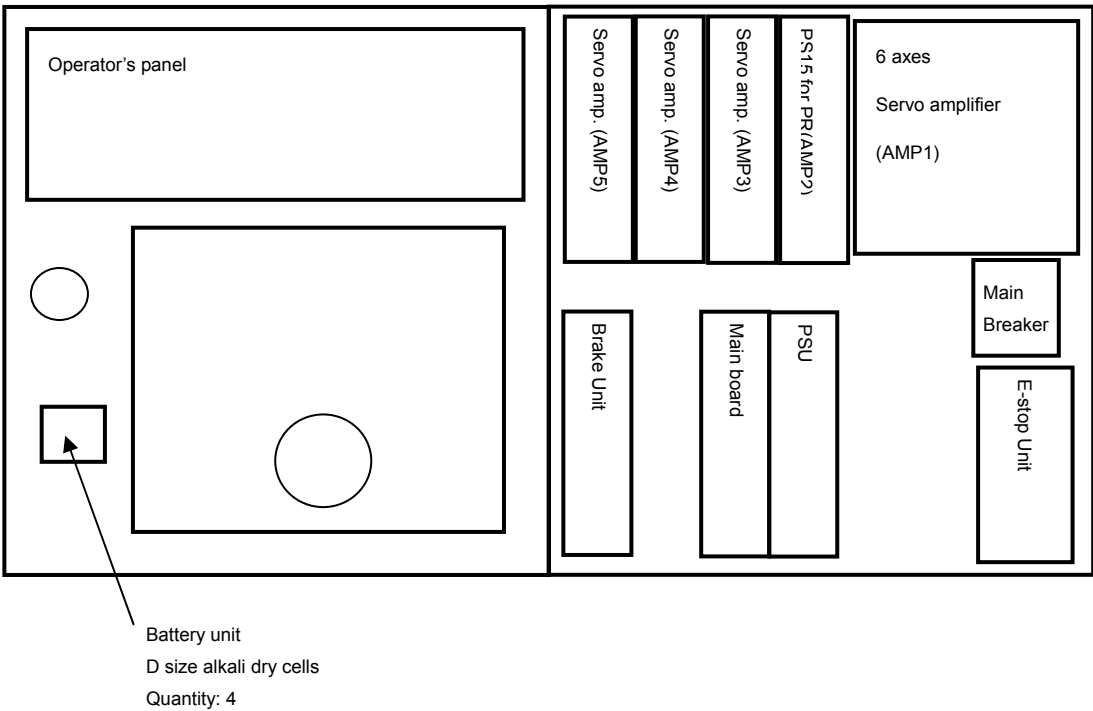
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2.2.2. In case of Discharge resistor B(2 elements/2000W, A05B-2603-H062)

Robot: R-2000iB/100P, M-410iB, R-2000iB/200T, R-2000iB/220U, M-900iA

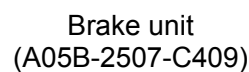
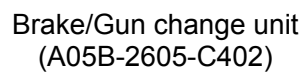
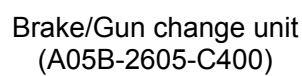


2.2.3. In case of Power regeneration



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3.1. Brake unit / Gun-change unit



Unit part number	Ordering specification including the unit
A05B-2605-C400	A05B-2605-J400
A05B-2605-C402	A05B-2607-J400
A05B-2507-C409	A05B-2605-J410
	A05B-2607-J410
	A05B-2607-J412

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4. Troubleshooting

4.1. Troubleshooting using the error code

4.1.1. SRVO-021 SRDY off (Group:i Axis:j)

[Explanation] When HDRY is on, SRDY is off even though no other alarm or emergency stop condition is present. This alarm can also occur when HRDY is off and the servo amplifier MCC is ON. HRDY is the signal sent from the host (Main board) to the servo system to specify whether to turn the servo amplifier MCC (located on E-Stop Unit) on or off. SRDY is the signal sent from the servo system to the host to indicate whether the servo amplifier MCC is on or off. If the servo amplifier MCC cannot be turned on when directed so, it is likely that a servo amplifier alarm has occurred. The host does not issue this alarm (SRDY off) if an alarm for the servo amplifier is detected. Therefore, this alarm indicates that the MCC cannot be turned on for an unknown reason.

Check the alarm history and investigate the factor of this alarm as follows.

<Case 1> If this alarm occurs on the 6ch-amplifier and the auxiliary axis servo amplifiers at the same time,

[Action 1] It is likely that the supply voltage stops instantaneously. Check the power supply.

[Action 2] Replace the E-stop unit.

[Action 3] Replace the 6ch-amplifier.

<Case 2> If this alarm occurs with other alarms,

[Action 1] In case other alarms occurs at the same time in the alarm history, check the troubleshooting of other alarms.

[Action 2] In case with "SRVO-214 Fuse Blown (Amp)", see the maintenance manual of R-30iB controller.

[Action 3] In case with "SRVO-222 Lack of Amp (Amp:i)", see the troubleshooting of SRVO-222.

<Case 3> If this alarm occurs on only the auxiliary axis servo amplifiers,

[Action 1] Replace the auxiliary axis servo amplifier that is indicated by the alarm message.

[Action 2] Replace the 6ch-amplifier.

<NOTE> A SRDY off may occur when the software setting is incorrect.

<1> If the parameter of the auxiliary axis is wrong, SRDY off occurs.

<2> If the machine lock is set to the 6ch-amplifier, SRDY off occurs on the auxiliary axis that is subordinate to the 6ch-amplifier.

<3> If the machine lock is set to one axis of the 2-axis and 3-axis servo amplifier, SRDY off occurs on the other auxiliary axis.

4.1.2. SRVO-022 SRDY on (Group:i Axis:j)

[Explanation] After inputting the reset signal, the ready signal from the servo amplifier was on before outputting the signal that let the amplifier to be ready.

[Action 1] Replace the servo card or the auxiliary axis board indicated by the alarm message.

[Action 2] Replace the servo amplifier that is indicated by the alarm message.

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4.1.3. SRVO-023 Stop error excess (Group:i Axis:j)

[Explanation] The position error becomes abnormally large, when the servo is at stop,

[Action 1] Check whether the robot or the auxiliary axis has collided with anything.

[Action 2] Check whether the motor brake has been released.

<1> Check the continuity of the brake circuit (cable, coil) at the CRR65*.

<2> Make sure that the brake cable has no ground fault.

<3> In case the brake unit is mounted, check to see if the fuse (FU2) of the brake unit is blown.(If this fuse is blown, it may be caused by some defect of the brake cable.)

[Action 3] Make sure that CZ2 connector at the servo amplifier and the power connector at the motor are connected tightly and check the continuity of the servo power circuit at CZ2.

[Action 4] Check to see if the load is greater than the rating. If greater, reduce it to within the rating. (If the load is too great, the torque required for acceleration / deceleration becomes higher than the capacity of the motor. As a result, the motor becomes unable to follow the command, and an alarm is issued.

[Action 5] Check to see if the input voltage of the controller is within its rated voltage and the voltage setting of the transformer is right. (If the voltage input to the servo amplifier becomes low, the torque output also becomes low. As a result the motor may become unable to follow the command, hence possibly an alarm.)

[Action 6] Replace the servo amplifier indicated by the alarm message.

[Action 7] Replace the brake unit.

[Action 8] Replace the motor.

4.1.4. SRVO-024 Move error excess (Group:i Axis:j)

[Explanation] When the robot is running, its position error is greater than a specified value (\$PARAM_GROUP.\$MOVER_OFFST or \$PARAM_GROUP.\$TRKERRLIM). It is likely that the motor cannot follow the speed specified by program.

[Action] Take the same actions as for SRVO-023.

4.1.5. SRVO-036 Inpos time over (Group:i Axis:j)

[Explanation] The robot did not get to the specified area (\$PARAM_GROUP.\$INPOS TIME) even after the position check monitoring time (\$PARAM_GROUP.\$STOPTOL) elapsed.

[Action] Take the same actions as for SRVO-023.

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4.1.6. SRVO-038 Pulse mismatch (Group:i Axis:j)

[Explanation] The pulse count stored when power was turned off does not match the count obtained when the power is turned on. This alarm will be asserted after changing the pulse coder, changing the back up battery for the pulse coder data or after loading back up data into the main board. Check the alarm history and investigate the factor of this alarm as follows.

[Action 1] In case with “SRVO-222 Lack of Amp (Amp:i)”, see the troubleshooting of SRVO-222.

[Action 2] If the brake number is set to the non-brake motors, this alarm may occur. Check the software setting of the brake number.

[Action 3] In case the robot or the auxiliary axis motors have been moved using the brake release unit while the power is off or when restoring the back-up data to the main board, this alarm may occur. Remaster the robot and the auxiliary axis motors.

[Action 4] If the robot or the auxiliary axis motors have been moved because the brake failed, this alarm may occur. Remove the factors and remaster the robot and the auxiliary axis motors.

[Action 5] Replace the pulse coder and remaster the group axis that is indicated by the alarm message.

4.1.7. SRVO-045 HVAL alarm (Group:i Axis:j)

[Explanation] Abnormally high current flowed in the main circuit of the servo amplifier.

[Action 1] Turn off the power, and disconnect the power cable for the auxiliary axis motor from the servo amplifier indicated by the alarm message.(and disconnect the brake cable to avoid the axis falling unexpectedly.) Supply power and see if the alarm occurs again. If the alarm occurs again, replace the servo amplifier.

[Action 2] Turn off the power and disconnect the power cable for the auxiliary axis(CZ2*) indicated by the alarm message, and check the insulation of their U, V, W and the GND lines each other. If there is a short-circuit, replace the power cable.

[Action 3] Turn off the power and disconnect the power cable for the auxiliary axis (CZ2*) indicated by the alarm message, and measure the resistance between their U and V,V and W and W and U with an ohmmeter that has a very low resistance range. If the resistances at the three places are different from each other, the motor, the power cable for auxiliary axis is defective. Check each item in detail and replace it if necessary.

4.1.8. SRVO-046 OVC alarm (Group:i Axis:j)

[Explanation] This alarm is issued to prevent the motor from thermal damage that might occur when the root mean square current calculated within the servo system is out of the allowable range.

[Action 1] Check the operating conditions for the motor and relax the service conditions if it is possible.

[Action 2] Take the same actions as for SRVO-023.

4.1.9. SRVO-47 LVAL alarm (Group:i Axis:j)

[Explanation] The control power supply voltage(+24V) supplied to servo amplifier is abnormally low.

[Action 1] Replace the servo amplifier that is indicated by the alarm message.

[Action 2] Replace the power supply unit.

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4.1.10. SRVO-50 Collision Detect alarm (Group:i Axis:j)

[Explanation] The disturbance torque estimated by the servo software is abnormally high. (A collision has been detected.)

[Action 1] Check the software setting for this option.

[Action 2] Take the same actions as for SRVO-023.

4.1.11. SRVO-51 CUER alarm (Group:i Axis:j)

[Explanation] The offset of the current feedback value is abnormally high.

[Action 1] Replace the servo amplifier that is indicated by the alarm message.

4.1.12. SRVO-55 FSSB com error 1 (Group:i Axis:j)

[Explanation] A communication error has been detected by the servo amplifier.

[Action 1] Replace the communication cable (Optical fiber) on the COP10B side of the servo amplifier indicated by the alarm message.

[Action 2] Replace the servo amplifier that is indicated by the alarm message.

[Action 3] Replace the servo amplifier between main board and the indicated servo amplifier.

[Action 4] Replace the servo card or the auxiliary axis board.

4.1.13. SRVO-56 FSSB com error 2 (Group:i Axis:j)

[Explanation] A communication error has been detected by the servo software.

[Action] Take the same actions as described for the above alarm.

4.1.14. SRVO-57 FSSB disconnect (Group:i Axis:j)

[Explanation] Communication was broken between the main board and the servo amplifier.

[Action 1] Check whether fuse F4 in the power supply unit has blown.

[Action 2] Check whether fuse FS1 in the servo amplifier has blown.

[Action 3] Check the communication cable (Optical fiber) between the main board and the 6ch-amplifier and between the 6ch-amplifier and the auxiliary axis servo amplifier. Replace it if it is faulty.

[Action 4] Replace the servo card or the auxiliary axis.

[Action 5] Replace the 6ch-amplifier and the auxiliary amplifier.

[Action 6] Make sure that there is no ground fault of +5V in the pulse coder cable (robot connection cable, robot internal cable and the auxiliary axis cable).

<Caution> Before continuing to the next step, perform a complete controller back up to save all your programs and settings. Failure to perform this could result in damage to equipment or lost data.

[Action 7] Replace the main board.

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- 4.1.15. SRVO-061 CKAL alarm (Group:i Axis:j)
- 4.1.16. SRVO-062 BZAL alarm (Group:i Axis:j)
- 4.1.17. SRVO-063 RCAL alarm (Group:i Axis:j)
- 4.1.18. SRVO-064 PHAL alarm (Group:i Axis:j)
- 4.1.19. SRVO-065 BLAL alarm (Group:i Axis:j)
- 4.1.20. SRVO-066 CSAL alarm (Group:i Axis:j)
- 4.1.21. SRVO-067 OHAL2 alarm (Group:i Axis:j)
- 4.1.22. SRVO-068 DTERR alarm (Group:i Axis:j)
- 4.1.23. SRVO-069 CRCERR alarm (Group:i Axis:j)
- 4.1.24. SRVO-070 STBERR alarm (Group:i Axis:j)
- 4.1.25. SRVO-071 SPHAL alarm (Group:i Axis:j)
- 4.1.26. SRVO-072 PMAL alarm (Group:i Axis:j)

Refer to the R-30iB controller maintenance manual.

4.1.27. SRVO-073 CMAL alarm (Group:i Axis:j)

[Explanation] It is likely that the pulse coder is abnormal or the pulse coder has malfunctioned due to noise.

[Action 1] Reinforce the earth of the flange of the motor that is indicated by the alarm message.

[Action 2] Replace the pulse coder that is indicated by the alarm message.

4.1.28. SRVO-074 LDAL alarm (Group:i Axis:j)

Refer to the R-30iB controller maintenance manual.

4.1.29. SRVO-076 Tip Stick Detection (Group:i Axis:j)

[Explanation] An excessive disturbance was assumed in servo software at the start of operation. (An abnormal load was detected. The cause may be welding.)

[Action 1] Take the same actions as for SRVO-023.

4.1.30. SRVO-135 FSAL alarm (Group:i Axis:j)

[Explanation] Faulty internal fan motor of the auxiliary axis servo amplifier.

[Action 1] Check whether there is any foreign material in the fan.

[Action 2] Be sure to push the faceplate (control printed-circuit board) as far as it will go.

[Action 3] Check that the fan connector is attached correctly.

[Action 4] Replace the fan.

[Action 5] Replace the auxiliary servo amplifier that is indicated by the alarm message.

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4.1.31. SRVO-136 DCLVAL alarm (Group:i Axis:j)

[Explanation] The DC link voltage is abnormally low.

-In case this alarm occurred during robot operation.

[Action 1] Check to see if the input voltage of the controller is within its rated voltage and the voltage setting of the transformer is right.

[Action 2] It is likely that the supply voltage stops instantaneously. Check the power supply.

[Action 3] Replace the E-STOP unit.

[Action 4] Replace the servo amplifier.

-In case this alarm occurs while the power up sequence or right after inputting reset signal:

[Action 1] Check whether the circuit breaker in the emergency stop unit is off. If it is off, check the servo amplifier and the wiring between the servo amplifier and the emergency stop unit. If anything is abnormal, replace it.

[Action 2] Check to see if the input voltage of the controller is within its rated voltage and the voltage setting of the transformer is right.

[Action 3] Replace the E-stop unit.

[Action 4] Replace the servo amplifier that is indicated by the alarm message.

-In case this alarm occurs right after pushing emergency stop button.

[Action 1] This alarm will occur in case the servo amplifier primary power is removed before an emergency stop. Check whether the auxiliary axis unit is installed correctly.

4.1.32. SRVO-156 IPMAL alarm (Group:i Axis:j)

[Explanation] Abnormally high current flowed through the main circuit of the servo power.

[Action] Take the same actions as for SRVO-045.

4.1.33. SRVO-215 Fuse blown (Aux. Axis)

[Explanation] The fuse (FU2) on the brake unit is blown.

[Action] Replace this fuse after replacing the brake cable because it may be defective. (If this fuse is blown, it may be caused by some defect of the brake cable.)

4.1.34. SRVO-222 Lack of Amp (Amp:i)

[Explanation] The servo amplifiers were not found at the power up sequence.

[Action 1] Turn off the controller, disconnect the pulse coder cable JF1,JF2 from the servo amplifier and turn on the controller again. If this alarm dose not occurs, the pulse coder cable may have +5V ground fault.(The LED of the servo amplifier may display blinking “-“ when the pulse coder cable has +5V ground fault.)

[Action 2] Replace the servo amplifier for auxiliary axis.

[Action 3] Replace the 6ch-amplifier.

[Action 4] Replace the optical fiber cable.

[Action 5] Replace the servo card or the auxiliary axis servo card.

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4.1.35. SRVO-251 DB relay abnormal (Group:i Axis:j)

[Explanation] An abnormality was detected in the internal relay (DB relay) of the servo amplifier.

[Action 1] Replace the auxiliary servo amplifier that is indicated by the alarm message.

4.1.36. SRVO-252 Current detect abnl (Group:i Axis:j)

[Explanation] An abnormality was detected in the current detection circuit inside the servo amplifier.

[Action 1] Replace the auxiliary servo amplifier that is indicated by the alarm message.

4.1.37. SRVO-290 DC link HC alarm (Group:i Axis:j)

[Explanation] The DC link current of the servo amplifier is abnormal

[Action] Take the same actions as for SRVO-045.

4.1.38. SRVO-291 IPM over heat (Group:i Axis:j)

[Explanation] The IPM in the servo amplifier overheated

[Action 1] Check the operating conditions for the motor and relax the service conditions if it is possible.

[Action 2] Check whether the amplifier fan has stopped.

[Action 3] If this alarm is frequency issued, replace the servo amplifier.

4.1.39. SRVO-292 EXT.FAN alarm (Group:i Axis:j)

[Explanation] The external fan of the auxiliary axis servo amplifier is faulty.

[Action 1] Check whether there is any foreign material in the fan.

[Action 2] Be sure to push the faceplate (control printed-circuit board) as far as it will go.

[Action 3] Check that the fan connector is attached correctly.

[Action 4] Replace the fan.

[Action 5] Replace the auxiliary servo amplifier that is indicated by the alarm message.

4.1.40. SRVO-295 SVM COM alarm (Group:i Axis:j)

[Explanation] A communication error occurred in the 6-axis amplifier or between the Power Supply(α iPS) and the servo amplifier.

[Action 1] Replace the 6-axis amplifier.

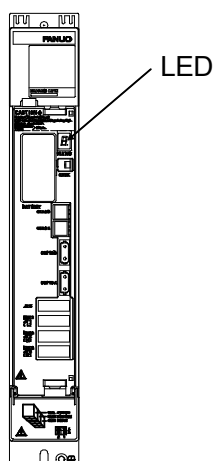
[Action 2] Replace the cable for communication the servo amplifier.

[Action 3] Replace the Power Supply(α iPS) if it is installed.

[Action 4] Replace the servo amplifier(α iSV)

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4.2. Troubleshooting based on indication.



4.2.1. Alarm Code 1

[Explanation] Inverter: internal cooling fan stopped

[Action 1] Check whether there is any foreign material in the fan.

[Action 2] Be sure to push the faceplate (control printed-circuit board) as far as it will go.

[Action 3] Check that the fan connector is attached correctly.

[Action 4] Replace the fan.

[Action 5] Replace the α iSV.

4.2.2. Alarm Code 2

[Explanation] Inverter: control power supply undervoltage

[Action 1] Check the three-phase input voltage of the amplifier (the voltage shall not be lower than 85% of the rated input voltage).

[Action 2] Check the 24 V power supply voltage output from the 6ch-amplifier (the voltage shall normally not lower than 22.8 V).

[Action 3] Check the connector and cable (CXA2A/B).

[Action 4] Replace the α iSV.

4.2.3. Alarm Code 5

[Explanation] Inverter: DC link undervoltage

[Action 1] Check that the screws for the DC link connection cable (bar) are tight.

[Action 2] If a DC link low voltage alarm condition occurs in one α iSV only, be sure to push the faceplate (control printed-circuit board) of that α iSV as far as it will go.

[Action 3] Replace the α iSV in which this alarm has occurred.

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4.2.4. Alarm Code 6

[Explanation] Inverter: overheat

- [Action 1] Check that the motor is being used at or below its continuous rating.
- [Action 2] Check that the cooling capacity of the cabinet is sufficient (inspect the fans and filters).
- [Action 3] Check that the ambient temperature is not too high.
- [Action 4] Be sure to push the faceplate (control printed-circuit board) as far as it will go.
- [Action 5] Replace the α iSV.

4.2.5. Alarm Code F

[Explanation] cooling fan stopped of the radiator

- [Action 1] Check whether there is any foreign material in the fan.
- [Action 2] Be sure to push the faceplate (control printed-circuit board) as far as it will go.
- [Action 3] Check that the fan connector is attached correctly.
- [Action 4] Replace the fan.
- [Action 5] Replace the α iSV.

4.2.6. Alarm Code P

[Explanation] Communication error between amplifier and module

- [Action 1] Check the connector and cable (CXA2A/B).
- [Action 2] Replace the control printed-circuit board.
- [Action 3] Replace the α iSV.

4.2.7. Alarm Code 8

[Explanation] Inverter: DC link current alarm

- [Action 1] Disconnect the motor power leads from the α iSV, and release the α iSV from an emergency stop condition.
 - <1> If no abnormal DC link current alarm condition has occurred
Go to [Action 2].
 - <2> If an abnormal DC link current alarm condition has occurred Replace the α iSV.
- [Action 2] Disconnect the motor power leads from the α iSV, and check the insulation between PE and the motor power lead U, V, or W.
 - <1> If the insulation is deteriorated
Go to [Action 3].
 - <2> If the insulation is normal
Replace the α iSV.
- [Action 3] Disconnect the motor from its power leads, and check whether the insulation of the motor or power leads is deteriorated.
 - <1> If the insulation of the motor is deteriorated
Replace the motor.

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<2> If the insulation of any power lead is deteriorated

Replace the power lead.

4.2.8. Alarm Codes 8., 9., and A.

(In case SRVO-156 IPMAL alarm (G:%d A:%d) occurs at the same time)

[Explanation] Inverter: IPM alarm

[Action 1] Be sure to push the faceplate (control printed-circuit board) as far as it will go.

[Action 2] Disconnect the motor power leads from the α SV, and release the α SV from an emergency stop condition.

<1> If no IPM alarm condition has occurred

Go to [Action 3].

<2> If an IPM alarm condition has occurred

Replace the α SV.

[Action 3] Disconnect the motor power leads from the α SV, and check the insulation between PE and the motor power lead U, V, or W.

<1> If the insulation is deteriorated

Go to [Action 4].

<2> If the insulation is normal

Replace the α SV.

[Action 4] Disconnect the motor from its power leads, and check whether the insulation of the motor or power leads is deteriorated.

<1> If the insulation of the motor is deteriorated

Replace the motor.

<2> If the insulation of any power lead is deteriorated

Replace the power lead.

4.2.9. Alarm Codes 8., 9., and A.

(In case SRVO-291 IPMAL overheat (G:%d A:%d) occurs at the same time)

[Explanation] Inverter: IPM alarm (OH)

[Action 1] Be sure to push the faceplate (control printed-circuit board) as far as it will go.

[Action 2] Check that the heat sink cooling fan is running.

[Action 3] Check that the motor is being used at or below its continuous rating.

[Action 4] Check that the cooling capacity of the cabinet is sufficient (inspect the fans and filters).

[Action 5] Check that the ambient temperature is not too high.

[Action 6] Replace the α SV.

4.2.10. Alarm Codes b, c, and d

[Explanation] Inverter: DC link current alarm

[Action 1] Check whether the servo parameters have default values. Alternatively, if an abnormal motor current

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alarm condition occurs only on rapid acceleration/deceleration, it is likely that the motor is being used under too harsh a condition. Increase the acceleration/deceleration time constant, and see what will occur.

[Action 2] Be sure to push the faceplate (control printed-circuit board) as far as it will go.

[Action 3] Disconnect the motor power leads from the α iSV, and release the α iSV from an emergency stop condition.

<1> If no abnormal motor current occurs

Go to [Action 4].

<2> If an abnormal motor current occurs

Replace the α iSV.

[Action 4] Disconnect the motor power leads from the α iSV, and check the insulation between PE and the motor power lead U, V, or W.

<1> If the insulation is deteriorated

Go to [Action 5].

<2> If the insulation is normal

Replace the α iSV.

[Action 5] Disconnect the motor from its power leads, and check whether the insulation of the motor or power leads is deteriorated.

<1> If the insulation of the motor is deteriorated

Replace the motor.

<2> If the insulation of any power lead is deteriorated

Replace the power lead.

4.2.11. Alarm Code "-" Blinking

[Explanation] Inverter: abnormal control power supply

[Action 1] Disconnect the feedback cable (JF*) from the α iSV, and then switch on the power.

<1> If blinking continues

Replace the α iSV.

<2> If blinking stops

Go to [Action 2].

[Action 2] Disconnect the feedback cable (JF*) from the Pulsecoder, and then switch on the power. (Keep the cable on the α iSV side connected.)

<1> If blinking continues

Replace the cable.

<2> If blinking stops

Replace the motor.

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4.2.12. Alarm Code U

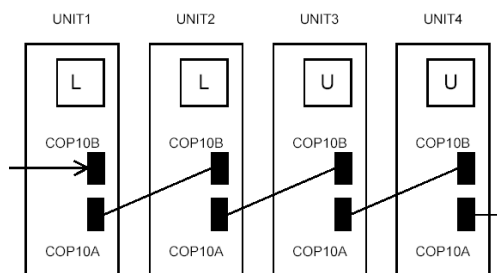
[Explanation] Inverter: FSSB communication error (COP10B) (NOTE)

[Action 1] Replace the α iSV optical cable (COP10B) that is nearest to the CNC on which "U" is displayed (the cable between UNIT2 and UNIT3).

[Action 2] Replace the α iSV that is nearest to the CNC on which "U" is displayed (UNIT3).

[Action 3] Replace the COP10B-side α iSV that is nearest to the CNC on which "U" is displayed (UNIT2).

[Action 4] Replace the servo card in the CNC.



NOTE

When the CNC power is turned on, "U" blinks momentarily, and then "-" steadily lights. This is not a failure, though.

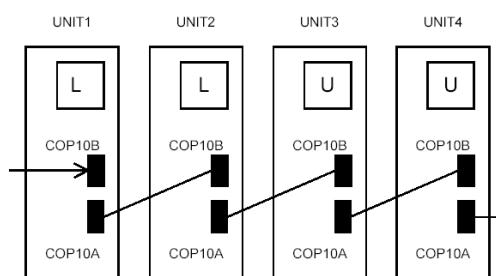
4.2.13. Alarm Code L

[Explanation] Inverter: FSSB communication error (COP10A)

[Action 1] Replace the α iSV optical cable (COP10A) that is farthest to the CNC on which "L" is displayed (the cable between UNIT2 and UNIT3).

[Action 2] Replace the α iSV that is farthest to the CNC on which "L" is displayed (UNIT2).

[Action 3] Replace the COP10A-side α iSV that is farthest to the CNC on which "L" is displayed (UNIT3).



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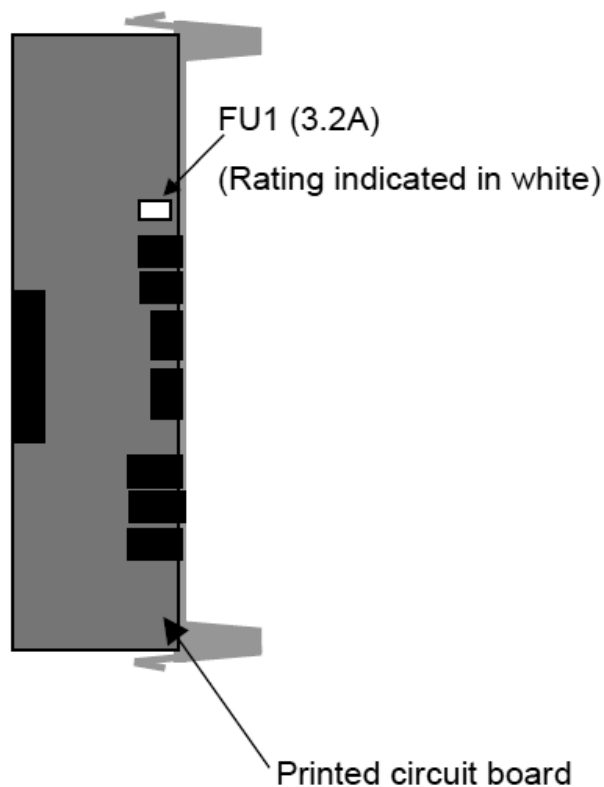
5. Replacing the parts

5.1. Replacing the fuse

5.1.1. Servo amplifier

There is one fuse to protect brake circuit on the servo amplifier.

Fuse (FU1) specification: A60L-0001-0290#LM32C



CAUTION

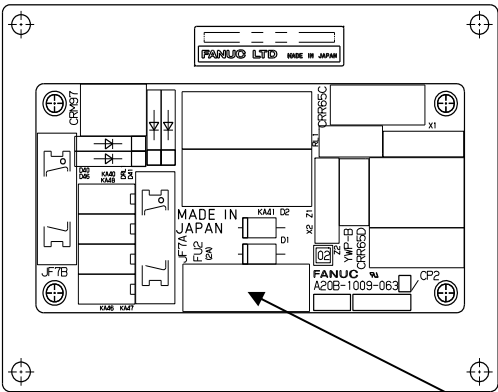
In case this fuse is blown, replace the amplifier because it is likely that this servo amplifier has some defect.

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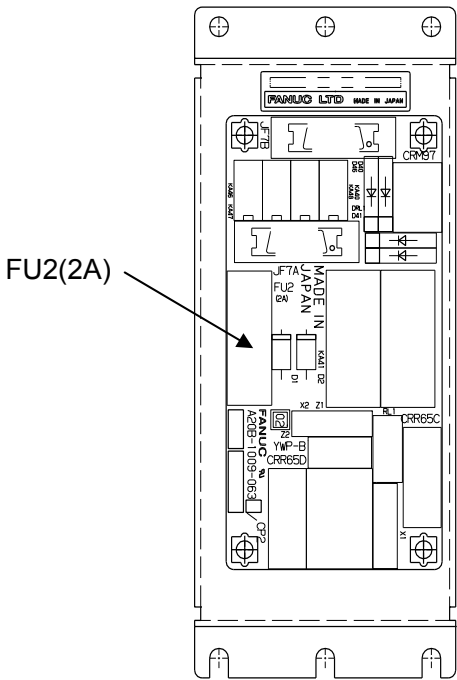
5.1.2. Brake/Gun change unit, Brake unit

There is one fuse to protect brake circuit on the brake/gun change unit or the brake unit.

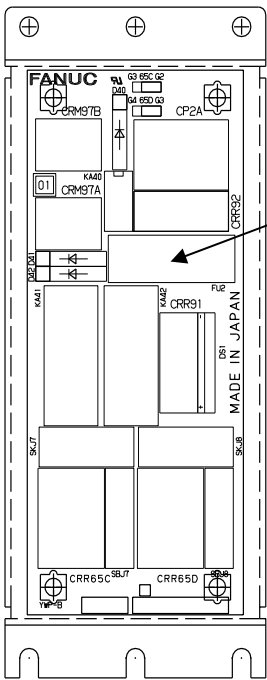
Fuse (FU2) specification: A60L-0001- 0245#GP20



brake/gun change unit
(A05B-2605-C400)



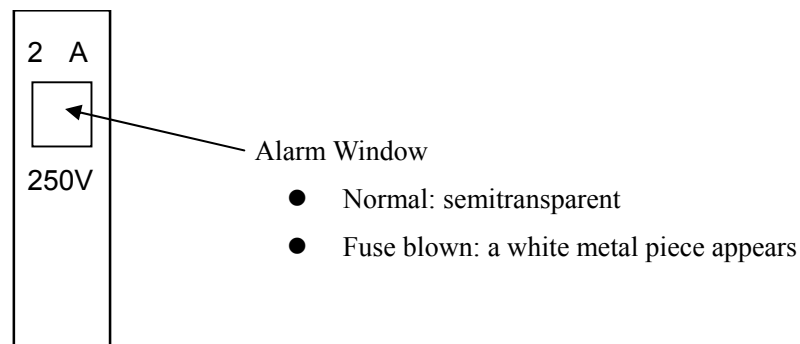
brake/gun change unit
(A05B-2605-C402)



brake unit
(A05B-2507-C409)

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This fuse has a window to indicate whether the fuse is blown or not. When the fuse is blown, a white metal piece appears on the window.



⚠ CAUTION

If this fuse is blown, brake circuit (brake cable for example) may have some defect. So, replace the fuse after repairing the failure.

5.2. Replacing the Dry cell

- If controller power is OFF, turn ON the controller power, and press an emergency stop switch.
- Remove the top panel of battery box.
- Do not make a mistake polarity when replace the dry cells.
- Close the top panel of battery box.

⚠ CAUTION

If you replace dry cells without controller power ON, the current position of motors will be lost.

⚠ CAUTION

Change the C or D size alkali dry cells (Q'ty: 4) every 18 months.

⚠ CAUTION

Some system does not have the battery unit for the auxiliary axis at the controller.

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5.3. Replacing the fan unit on αiSV

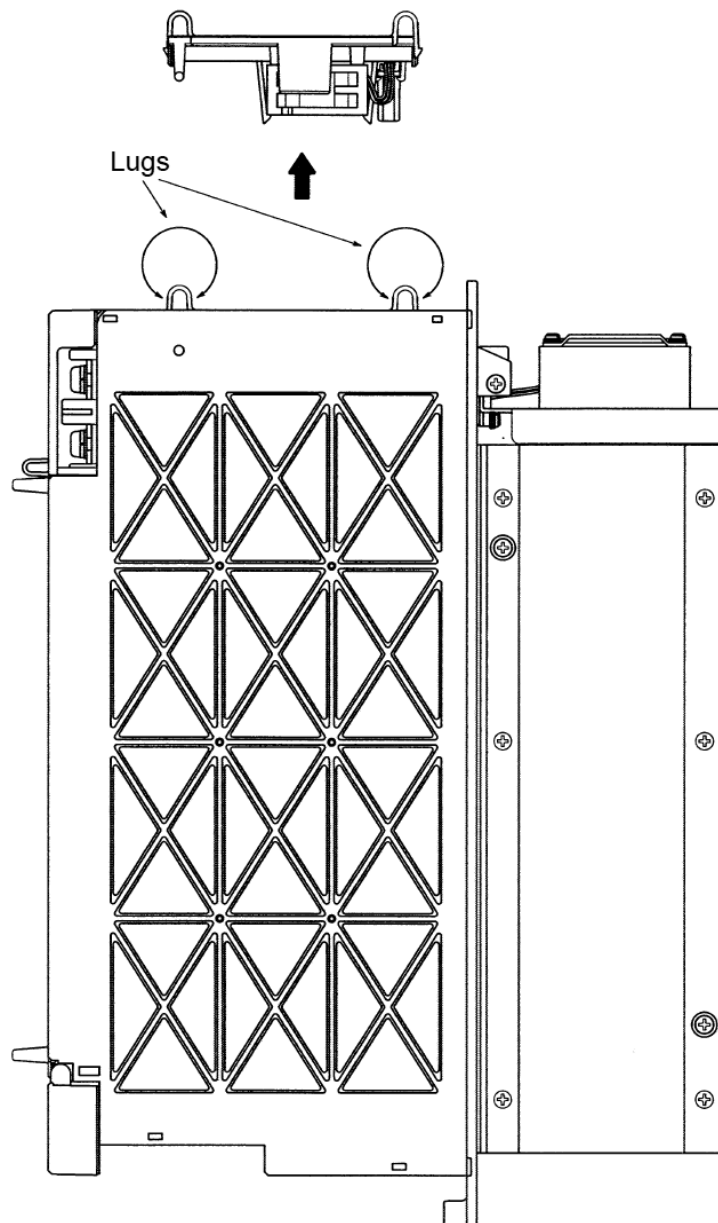
5.3.1. Fan for circulating the inside air

	Fan unit (*1)	Fan motor
αiSV 20	A06B-6110-C605	A90L-0001-0510
αiSV 40		
αiSV 80		
αiSV 160		
αiSV 20/20		
αiSV 20/40		
αiSV 40/40		
αiSV 40/80		
αiSV 80/80		
αiSV 80/160	A06B-6110-C606	
αiSV 160/160		
αiSV 20/20/20	A06B-6110-C605	
αiSV 20/20/40		

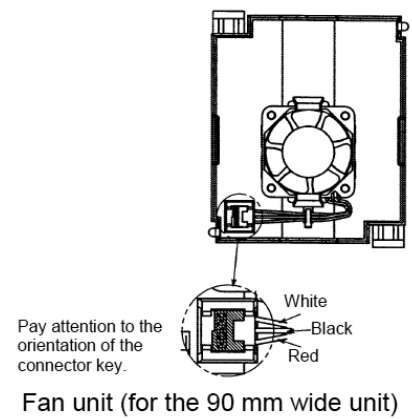
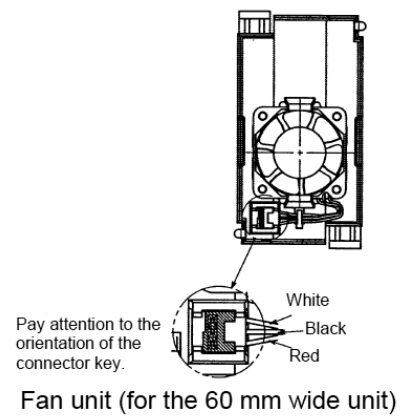
(*1) A fan unit is a set of a fan motor and a cover for mounting it. A fan motor can be replaced separately from the fan unit.

					Title			
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Holding the two lugs on the fan unit, lift the fan unit in the direction of the arrow (toward the right in the figure).



When replacing the fan motor, pay attention to its orientation and the orientation of the connector.



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5.3.2. Fan for cooling external heat sink fins

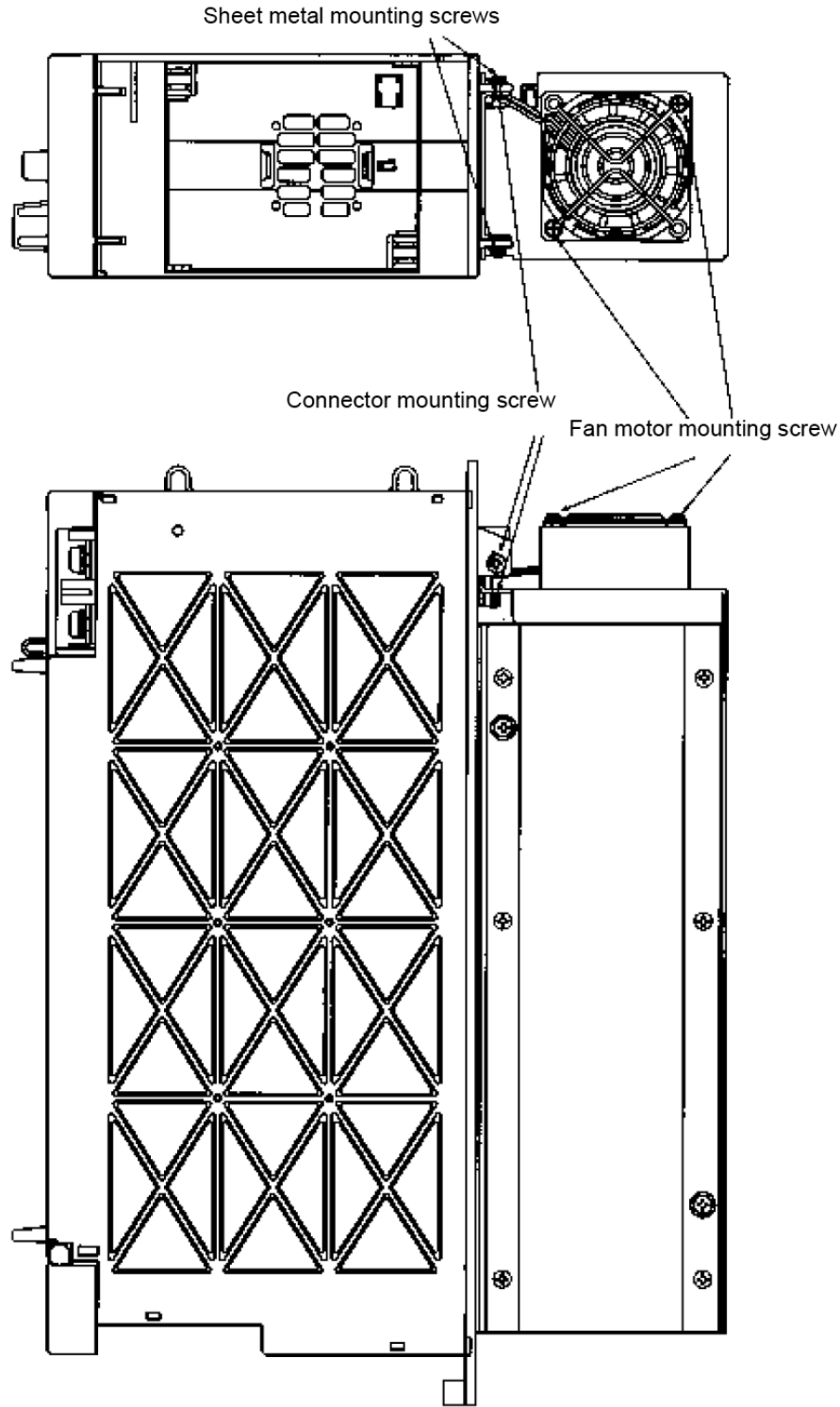
	Fan unit (*1)	Fan motor
αiSV 20	-	-
αiSV 40	-	-
αiSV 80	-	-
αiSV 160	A06B-6110-C602	A90L-0001-0507#B
αiSV 20/20	-	-
αiSV 20/40	-	-
αiSV 40/40	A06B-6110-C601	A90L-0001-0507#A
αiSV 40/80		
αiSV 80/80		
αiSV 80/160	A06B-6110-C603	A90L-0001-0508
αiSV 160/160		
αiSV 20/20/20	-	A90L-0001-0385#T (*2)
αiSV 20/20/40	-	-

(*1) A fan unit is a set of a fan motor and a cover for mounting it. A fan motor can be replaced separately from the fan unit.

(*2) The αiSV 20/20/20 has, within its housing, a fan motor for cooling the heat sink fins on the power transistor.

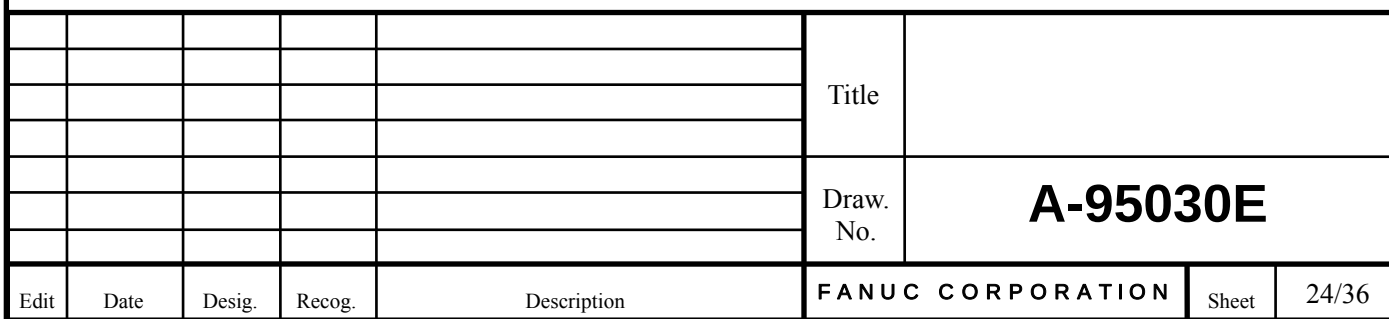
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					Draw. No.	A-95030E		
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- 1 Remove the sheet metal mounting screws (two), and detach the fan motor together with the sheet metal from the unit.
- 2 Remove the fan motor mounting screws (two for one fan motor and four for two fan motors).
- 3 Remove the connector mounting screws (two).

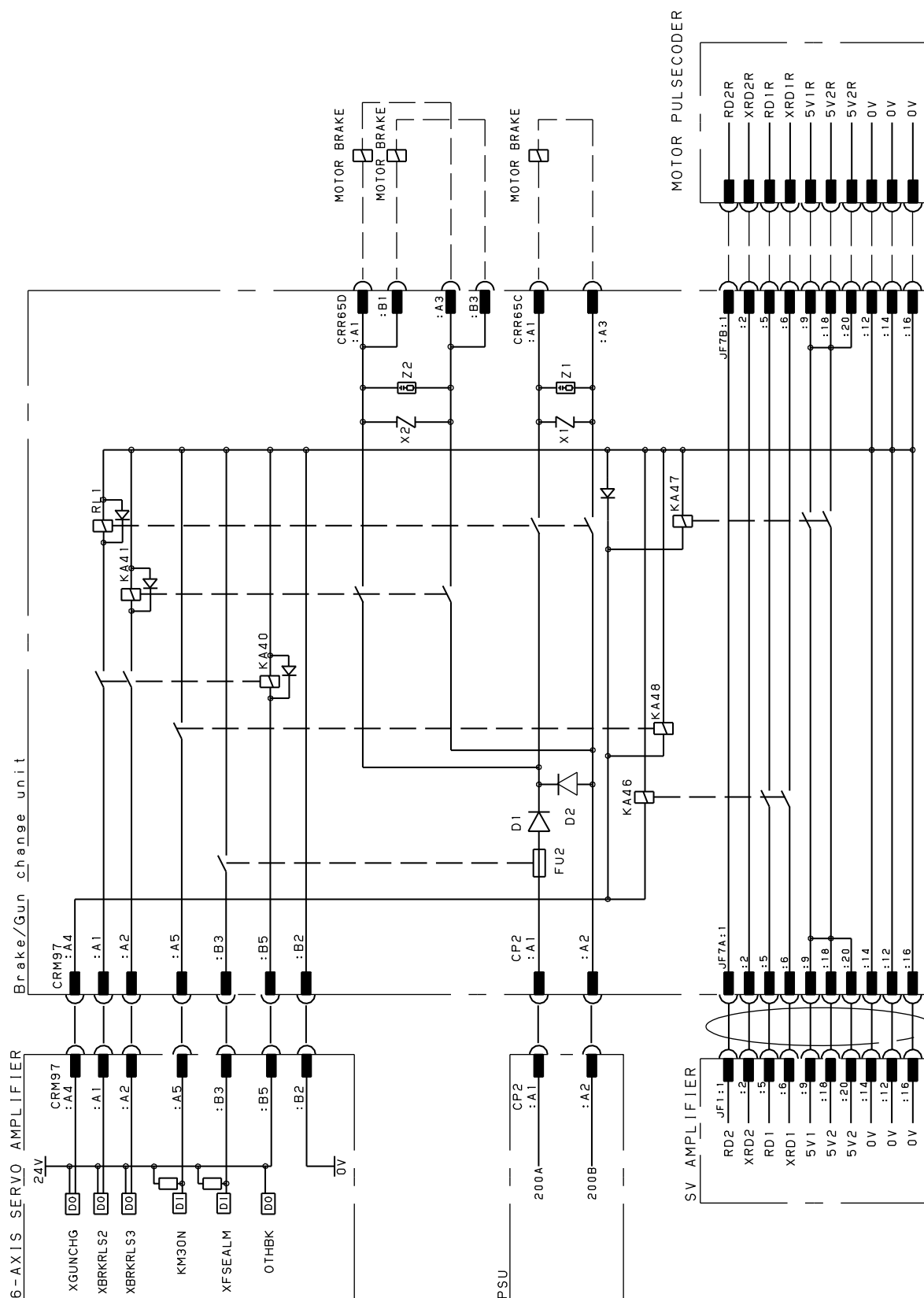


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					Draw. No.	A-95030E		
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6.1. Servo amplifier connection

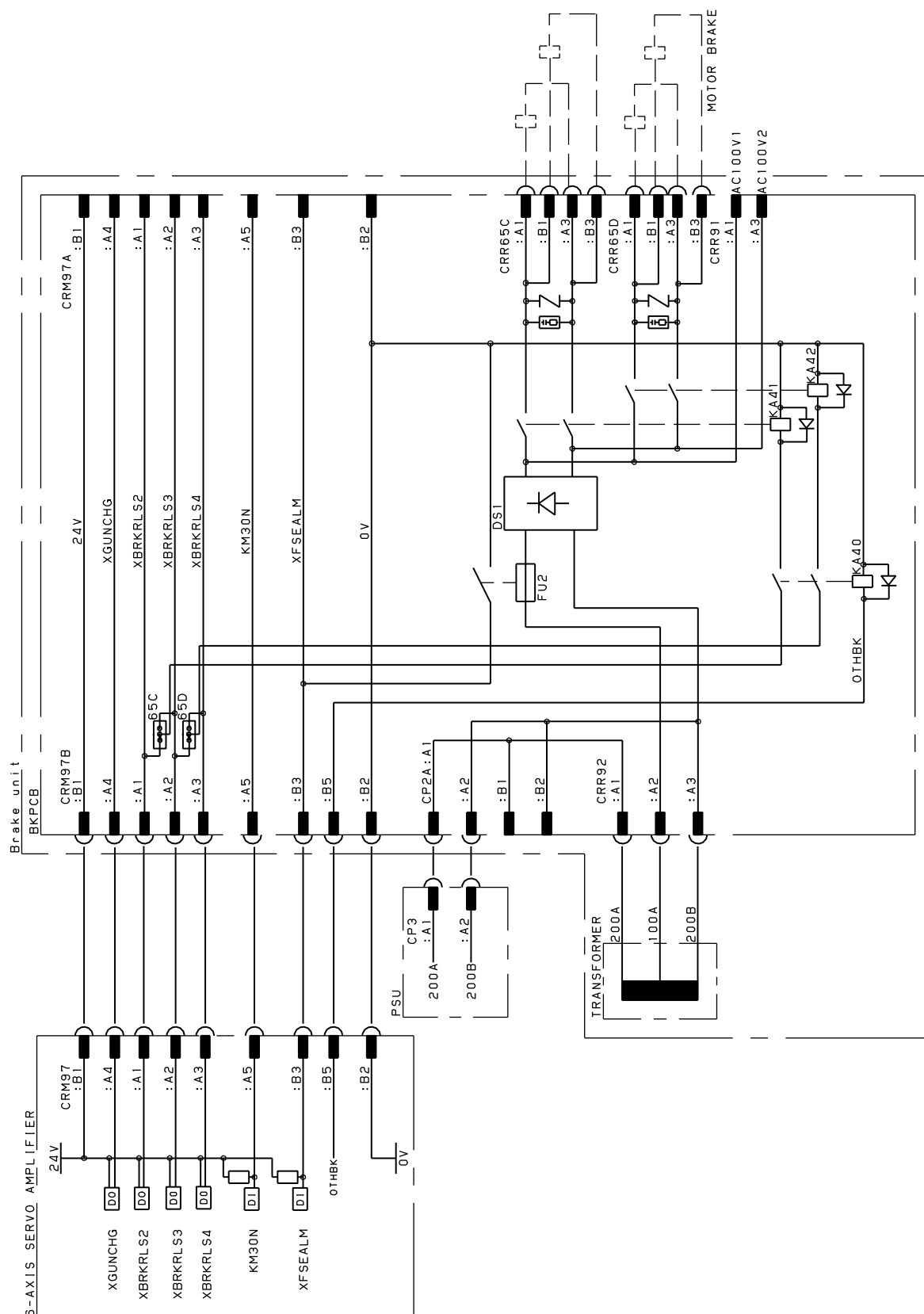


6.2. Brake / Gun change unit connection (A-cabinet) A05B-2605-J400



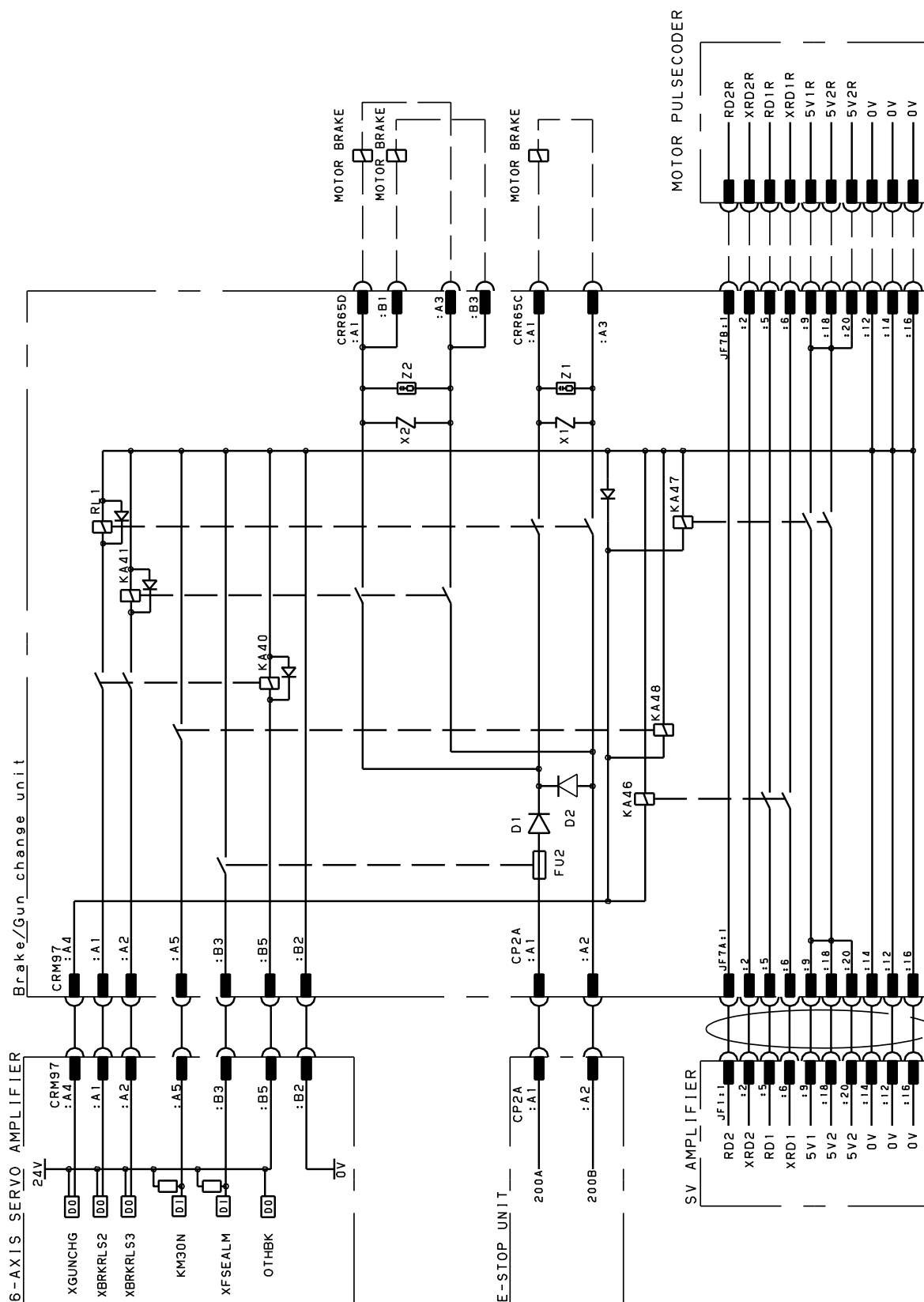
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6.3. Brake unit connection (A-cabinet) A05B-2605-J410



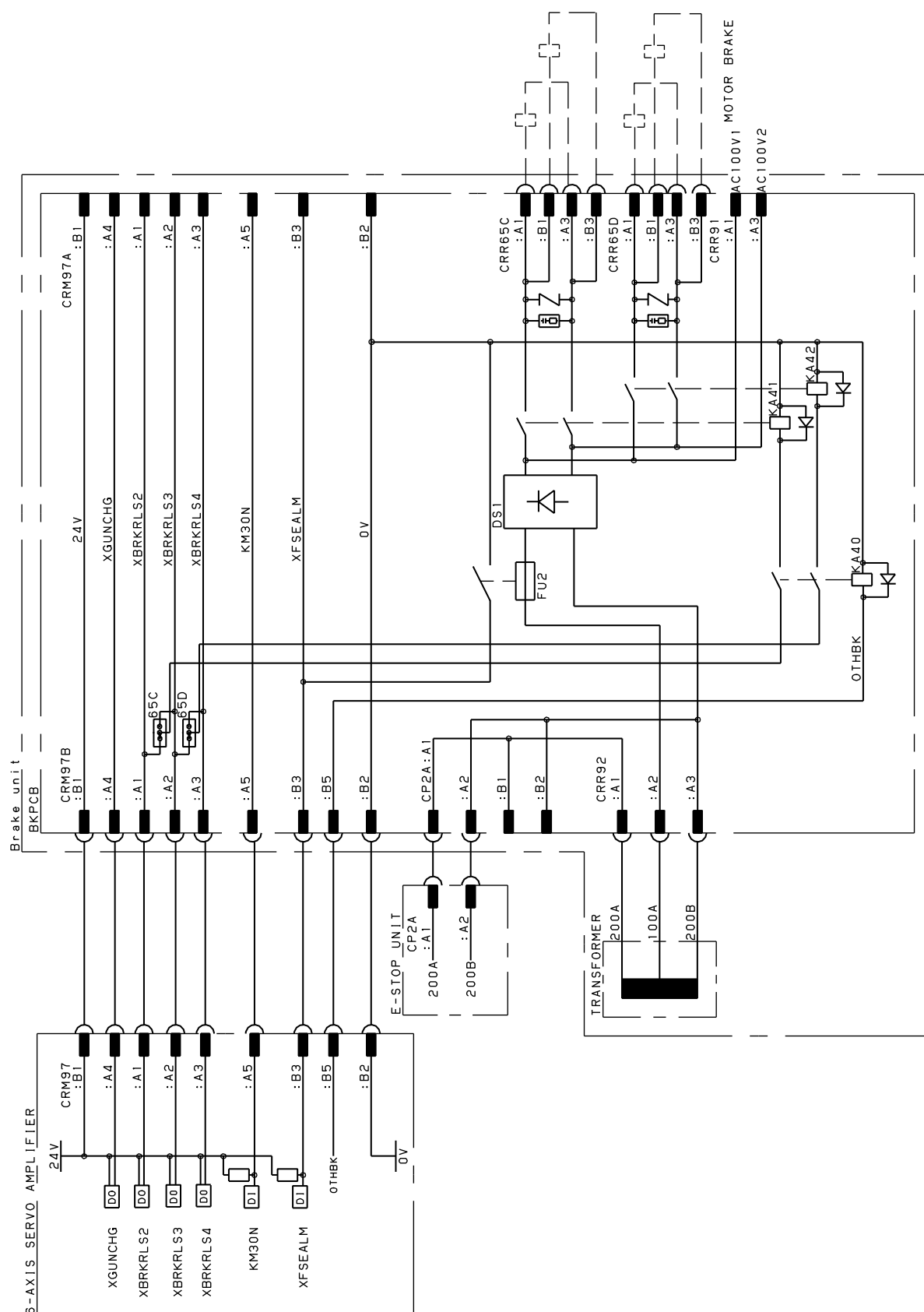
					Title	A-95030E		
					Draw. No.	A-95030E		
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6.4. Brake / Gun change unit connection (B-cabinet) A05B-2607-J400



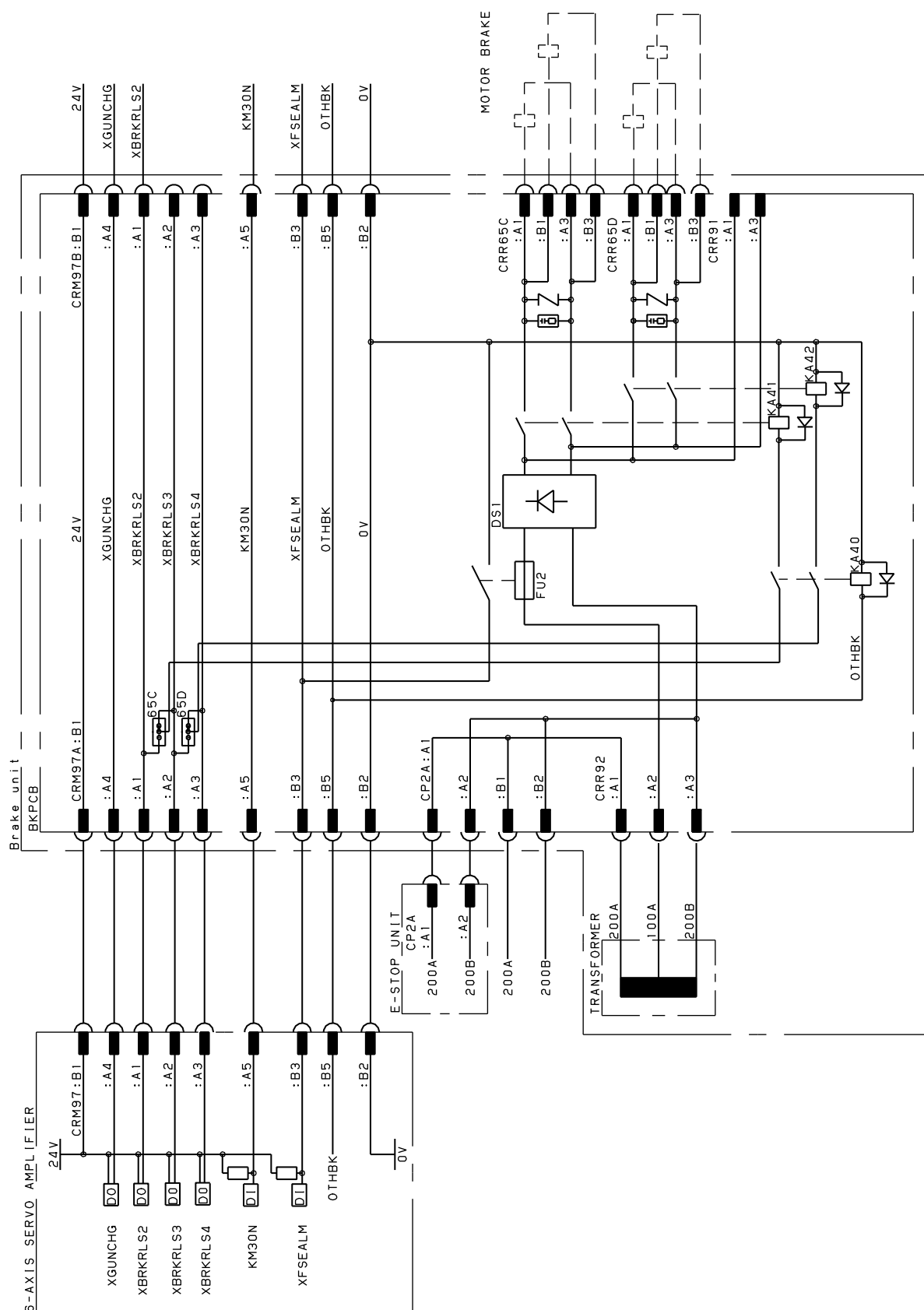
					Title	A-95030E		
					Draw. No.	A-95030E		
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6.5. Brake unit connection (B-cabinet/4-axes) A05B-2607-J410



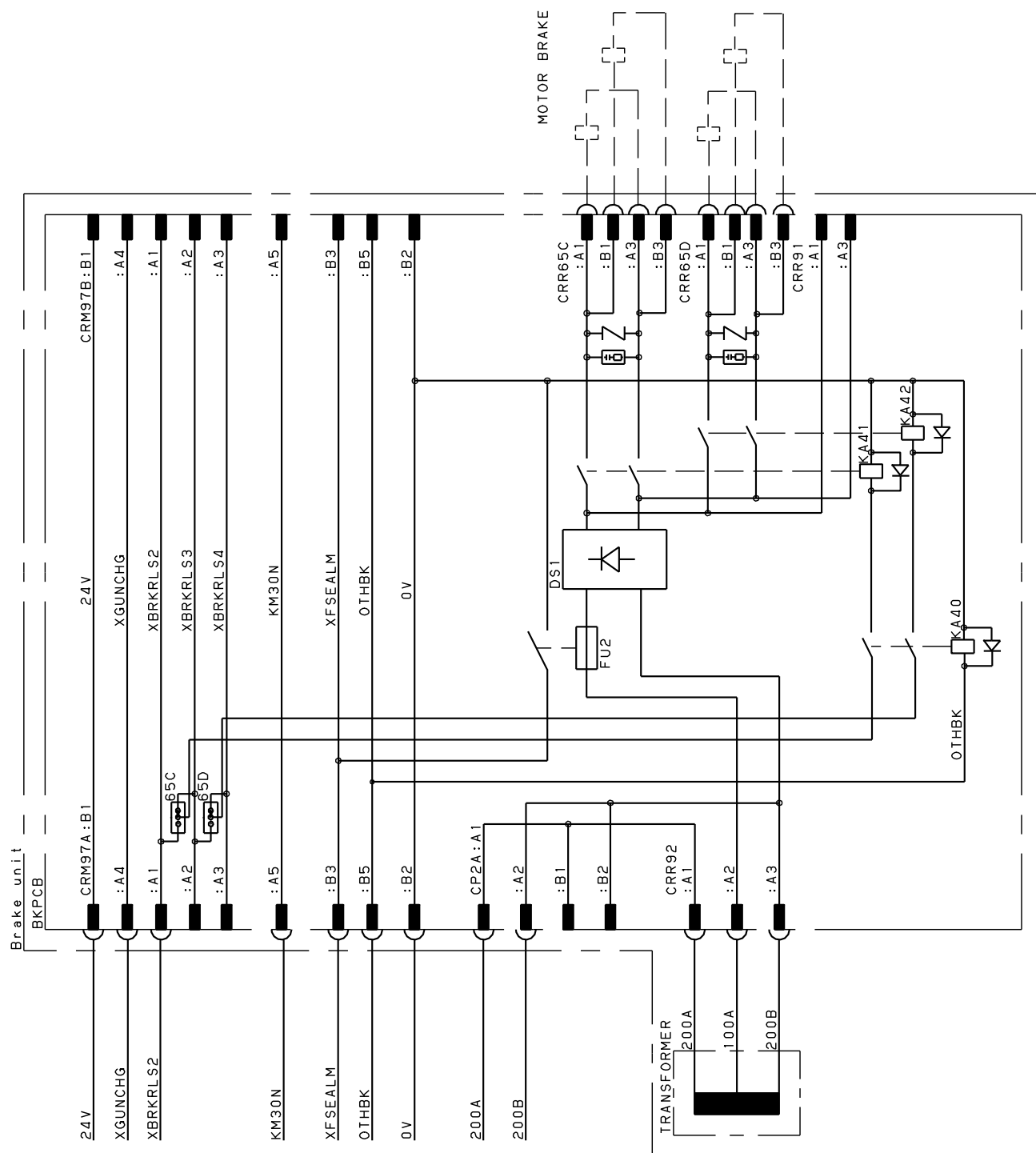
					Title	A-95030E		
					Draw. No.	A-95030E		
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6.6. Brake unit connection (B-cabinet/8-axes)(1/2) A05B-2607-J412



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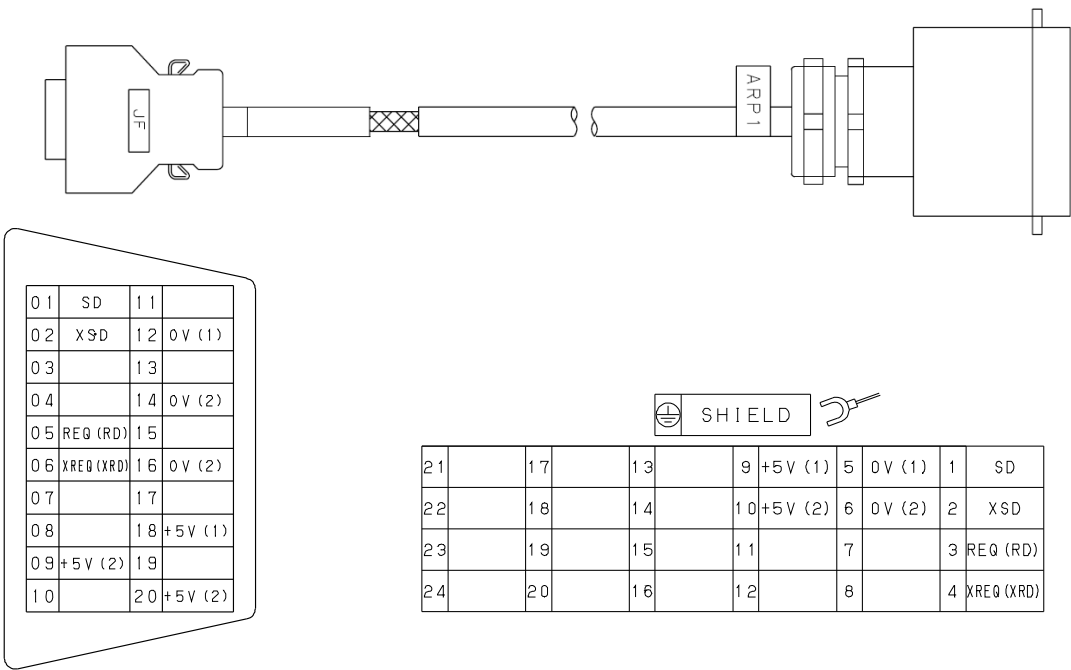
6.7. Brake unit connection (B-cabinet/8-axes)(2/2) A05B-2607-J412



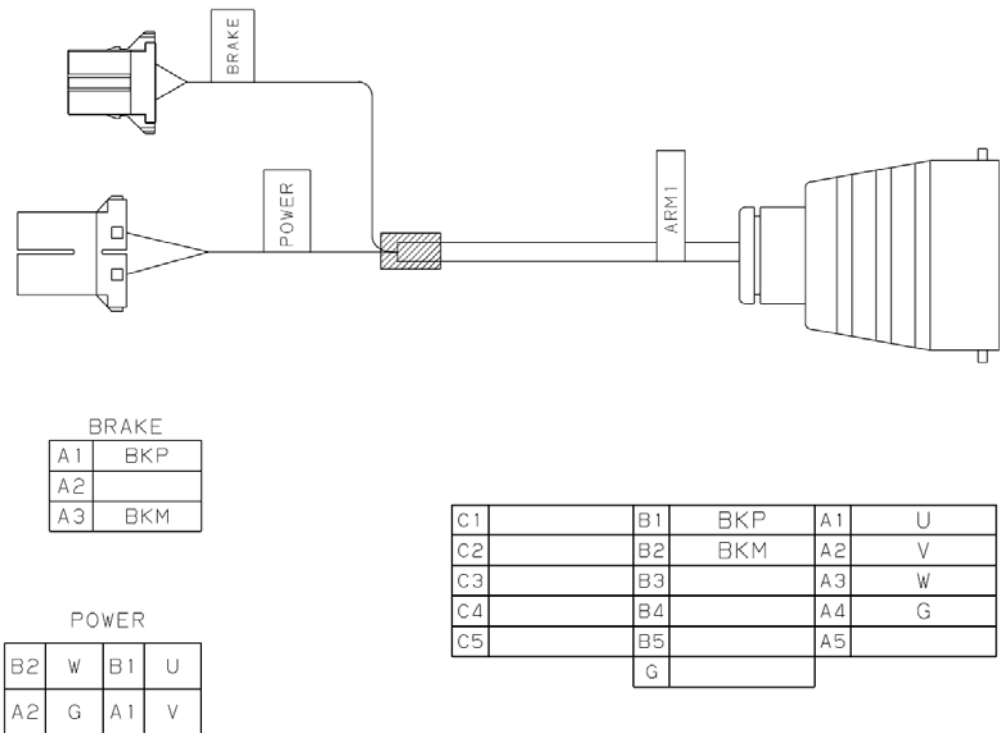
					Title	A-95030E		
					Draw. No.	A-95030E		
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7. Cable connection

7.1. ARP/J7P cable

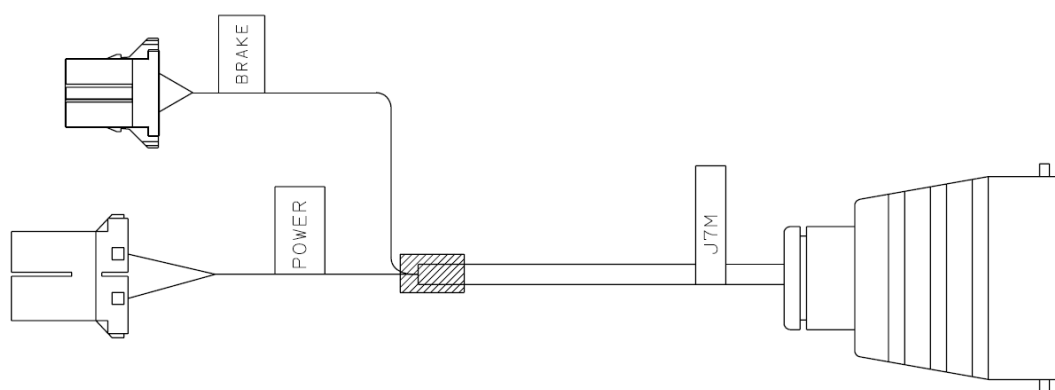


7.2. ARM cable



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7.3. J7M cable



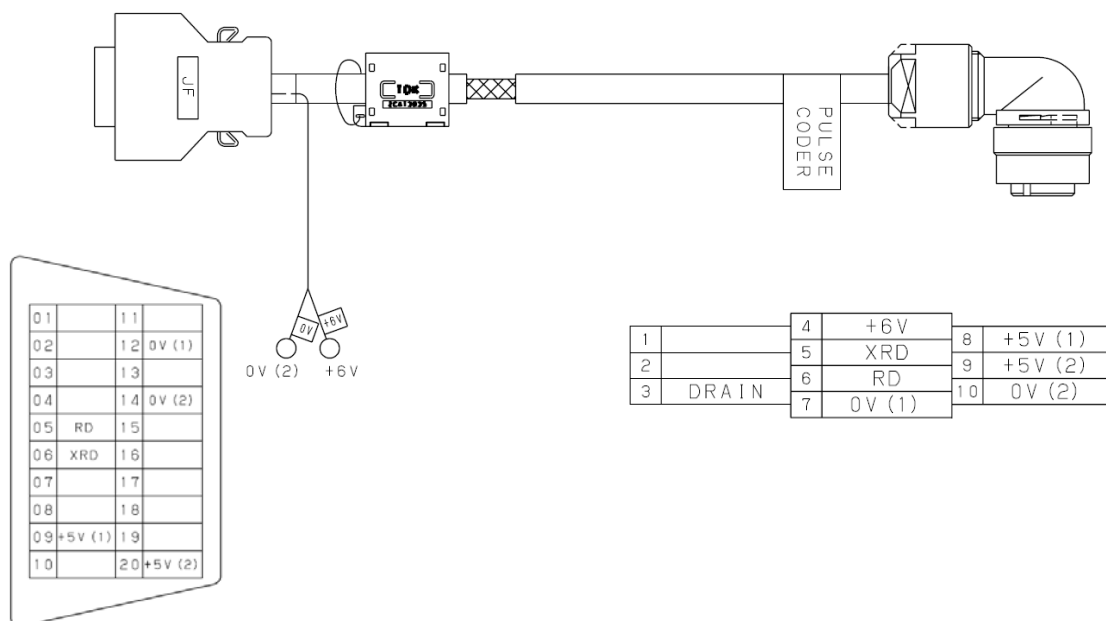
BRAKE	
A1	BKP
A2	
A3	BKM

POWER			
B2	W	B1	U
A2	G	A1	V

			
3	БКР	1	U
6	БКМ	4	V
9		7	W
12		10	G
		2	U *
		5	V *
		8	W *
		11	G *

*) In case of αiF30/3000,
αiF40/3000, αiS22/4000,
αiS30/4000, αiS40/4000

7.4. Pulse coder cable



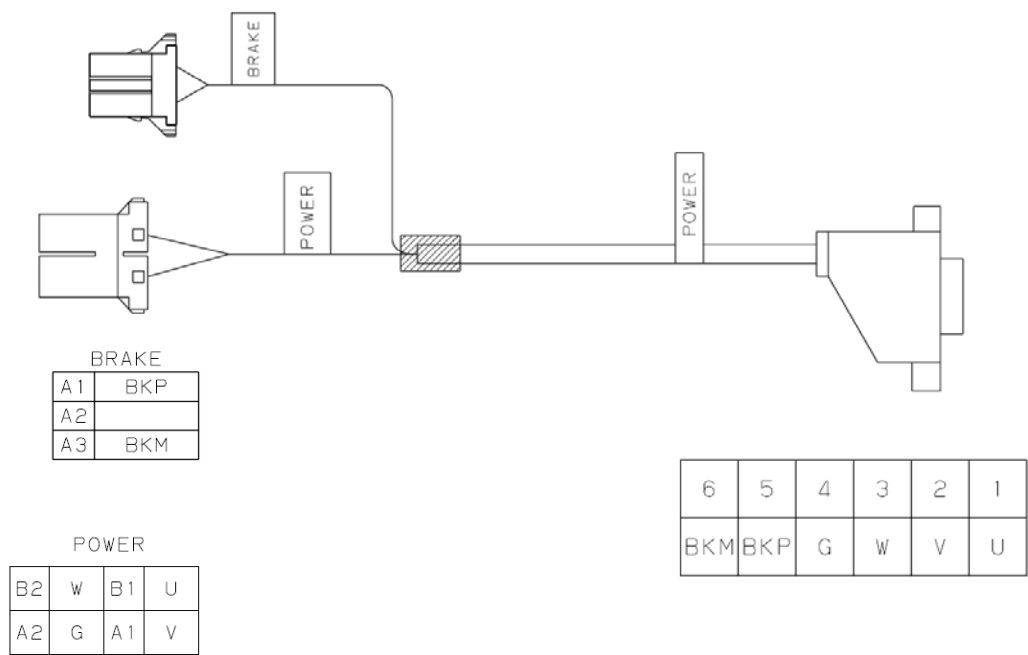
01		11	
02		12	0V (1)
03		13	
04		14	0V (2)
05	RD	15	
06	XRD	16	
07		17	
08		18	
09	+5V (1)	19	
10		20	+5V (2)

1		4	+6V	8	+5V (1)
2		5	XRD	9	+5V (2)
3	DRAIN	6	RD	10	0V (2)
		7	0V (1)		

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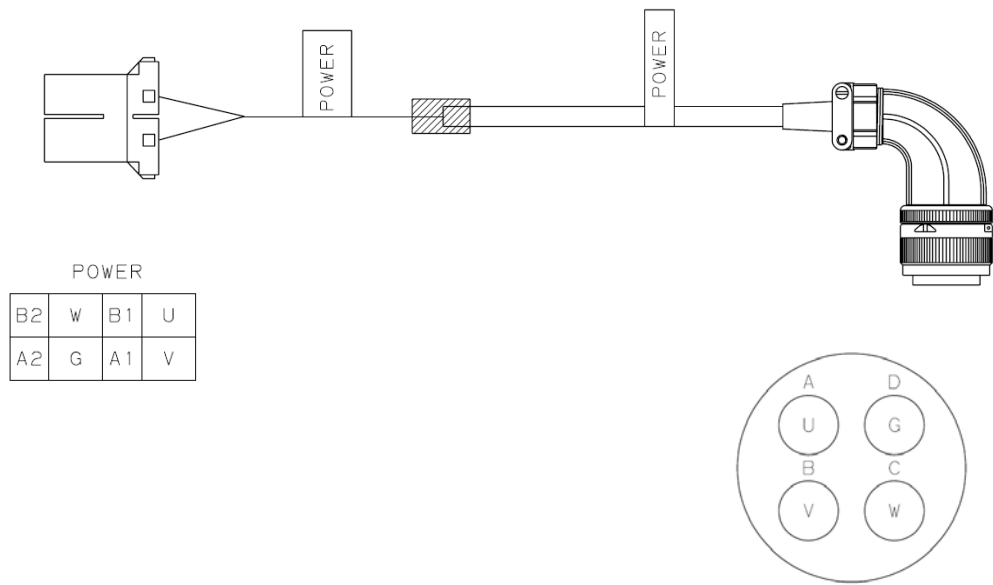
7.5. Motor power / Brake cable

For αiF1/5000, αiF2/5000, αiS2/5000, αiS4/5000, βiS2/4000, βiS4/4000



7.6. Motor power cable

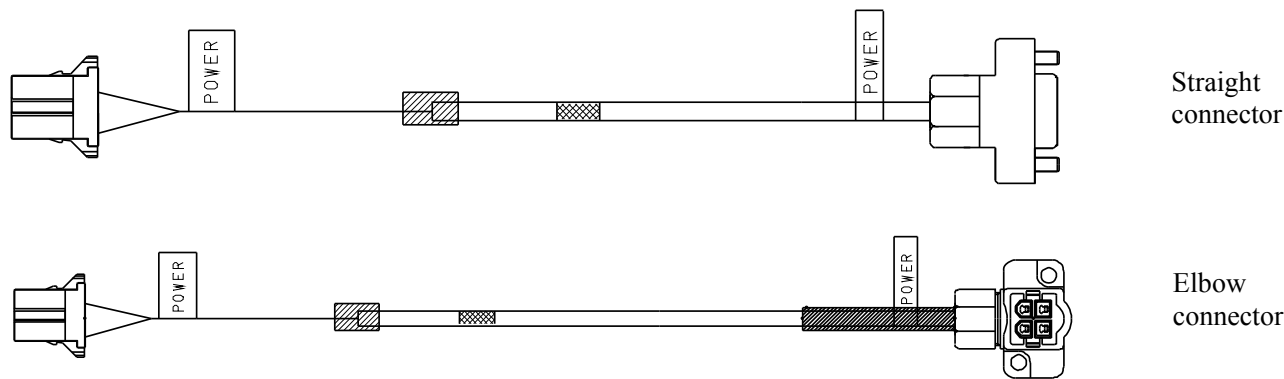
For αiF4/4000, αiF8/3000, αiS8/4000, αiS12/4000, βiS8/3000, βiS12/3000, αiF12/3000, αiF22/3000, βiS22/2000, αiF30/3000, αiF40/3000, αiS22/4000, αiS30/4000, αiS40/4000



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7.7. Motor Power cable

For β iS0.4/4000, β iS0.5/6000, β iS1/6000



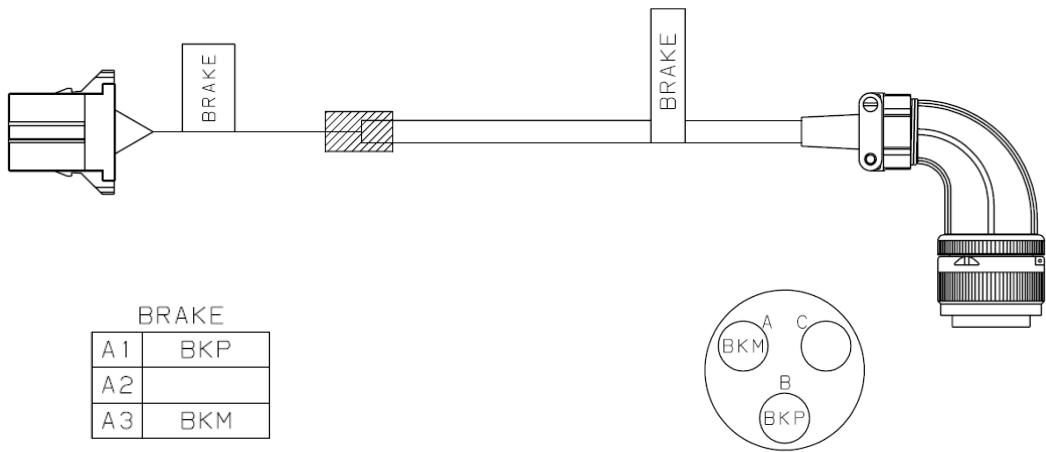
POWER

B2	W	B1	U
A2	G	A1	V

1	2
U	V
4	3
G	W

7.8. Motor Brake cable

For α iF4/4000, α iF8/3000, α iS8/4000, α iS12/4000, β iS8/3000, β iS12/3000, α iF12/3000, α iF22/3000, β iS22/2000, α iF30/3000, α iF40/3000, α iS22/4000, α iS30/4000, α iS40/4000



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For $\beta\text{iS}0.4/4000$, $\beta\text{iS}0.5/6000$, $\beta\text{iS}1/6000$



BRAKE	
A1	BKP
A2	
A3	BKM

1	2
	BKP
4	3
	BKM

JF 1			
01	PD1	11	
02	XPD1	12	0v1 {1}
03		13	
04		14	0v1 {2}
05	PRQ1	15	
06	XPRQ1	16	0v1 {2}
07		17	
08		18	+5v1 (1)
09	+5v1 {2}	19	
10		20	+5v1 {2}

JF 2			
01	PD2	11	
02	XPD2	12	0V2 (1)
03		13	
04		14	0V2 (2)
05	PRQ2	15	
06	XPRQ2	16	0V2 (2)
07		17	
08		18	+5V2 (1)
09	+5V2 (2)	19	
10		20	+5V2 (2)



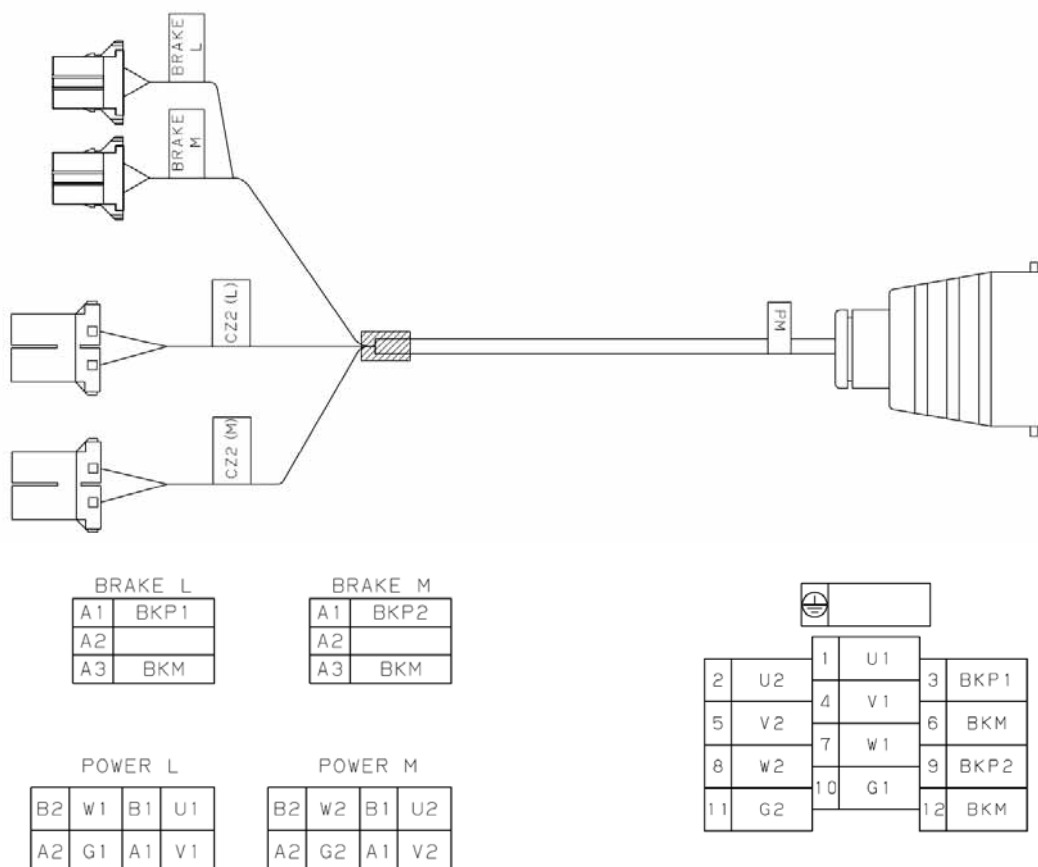
SHIELD



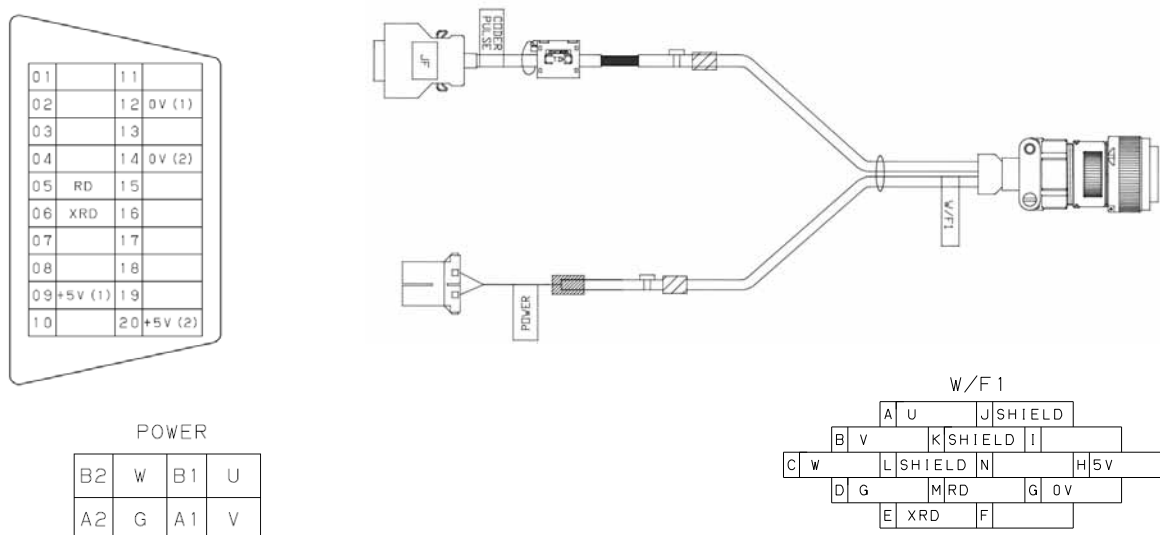
21	+5V2 (1)	17	0V2 (1)	13	PD2	9	+5V1 (1)	5	0V1 (1)	1	PD1
22	+5V2 (2)	18	0V2 (2)	14	XPD2	10	+5V1 (2)	6	0V1 (2)	2	XPD1
23		19		15	PRQ2	11		7		3	PRQ1
24		20		16	XPRQ2	12		8		4	XPRQ1

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7.11. PM cable



7.12. Serbo torch cable



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